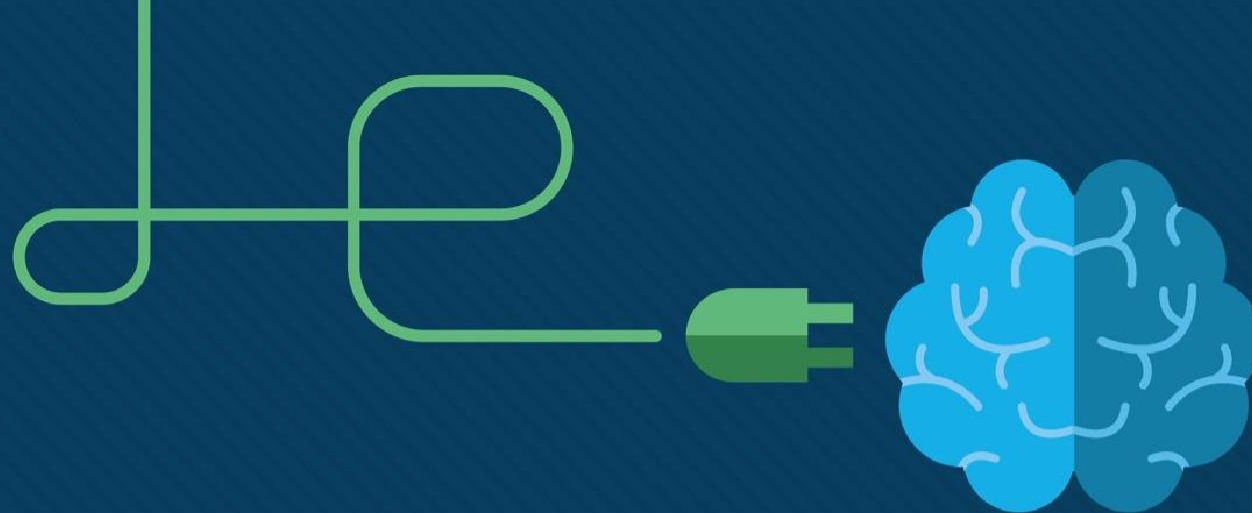




Chapter 8: Network Layer

Module 8: Network Layer Introduction
to Networks v7.02 (ITN)





Chapter 8 - Sections & Objectives

■ 8.1 Network Layer Protocols

- Explain how network layer protocols and services support communications across data networks
- Describe the purpose of the network layer in data communication.
- Explain why the IPv4 protocol requires other layers to provide reliability.
- Explain the role of the major header fields in the IPv4 packet.
- Explain the role of the major header fields in the IPv6 packet.

■ 8.2 Routing

- Explain how routers enable end-to-end connectivity in a small to medium-sized business network.
- Explain how network devices use routing tables to direct packets to a destination network.
- Compare a host routing table to a routing table in a router.

■ 8.3 Routers

- Explain how devices route traffic in a small to medium-sized business network
- Describe the common components and interface of a router.
- Describe the boot-up process of a Cisco IOS router.

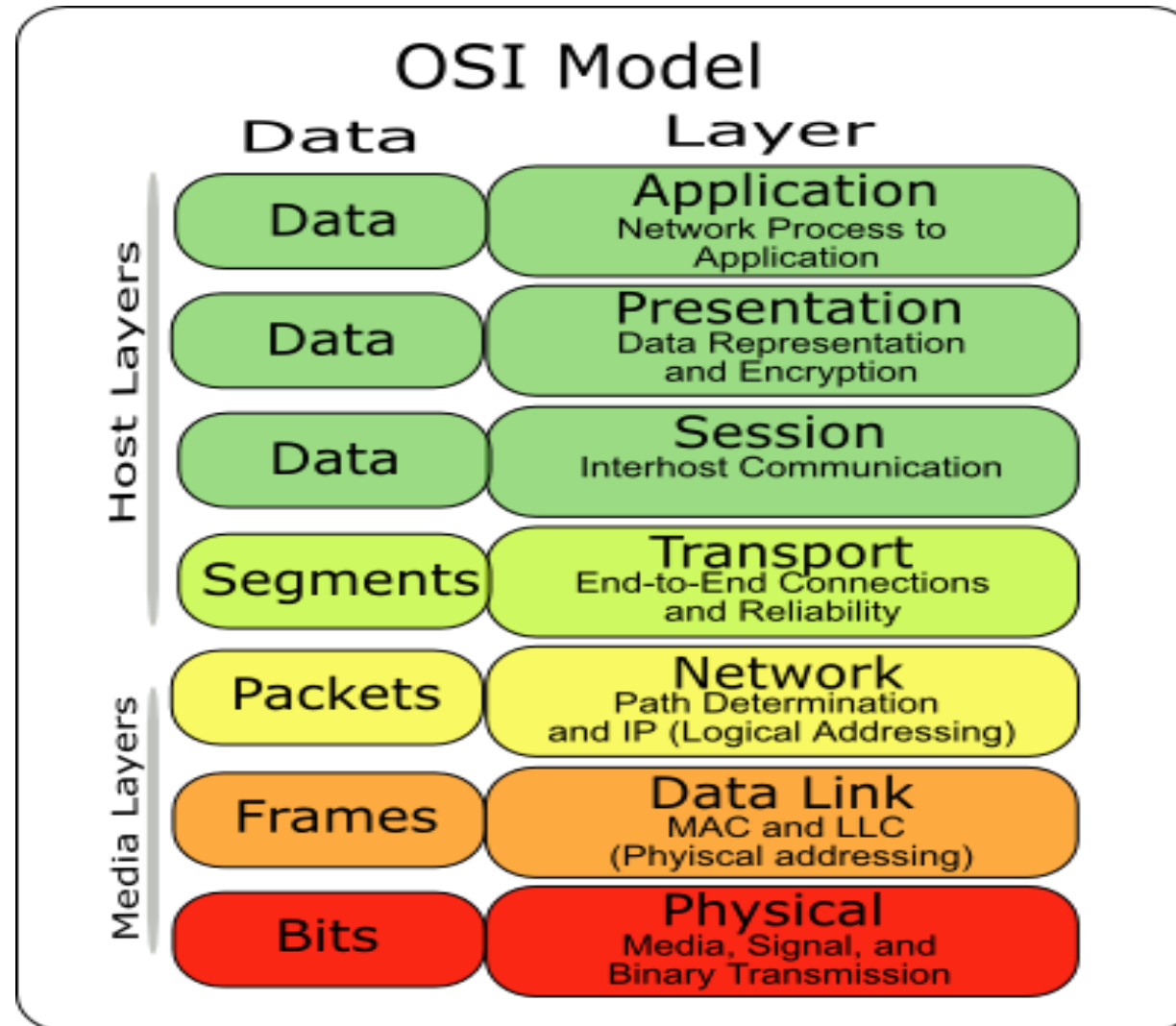
8.1 Network Layer Protocols

8.1 Network Layer Characteristics

Module 8: Network Layer



8.1 Network Layer Protocols





Network Layer in Communications

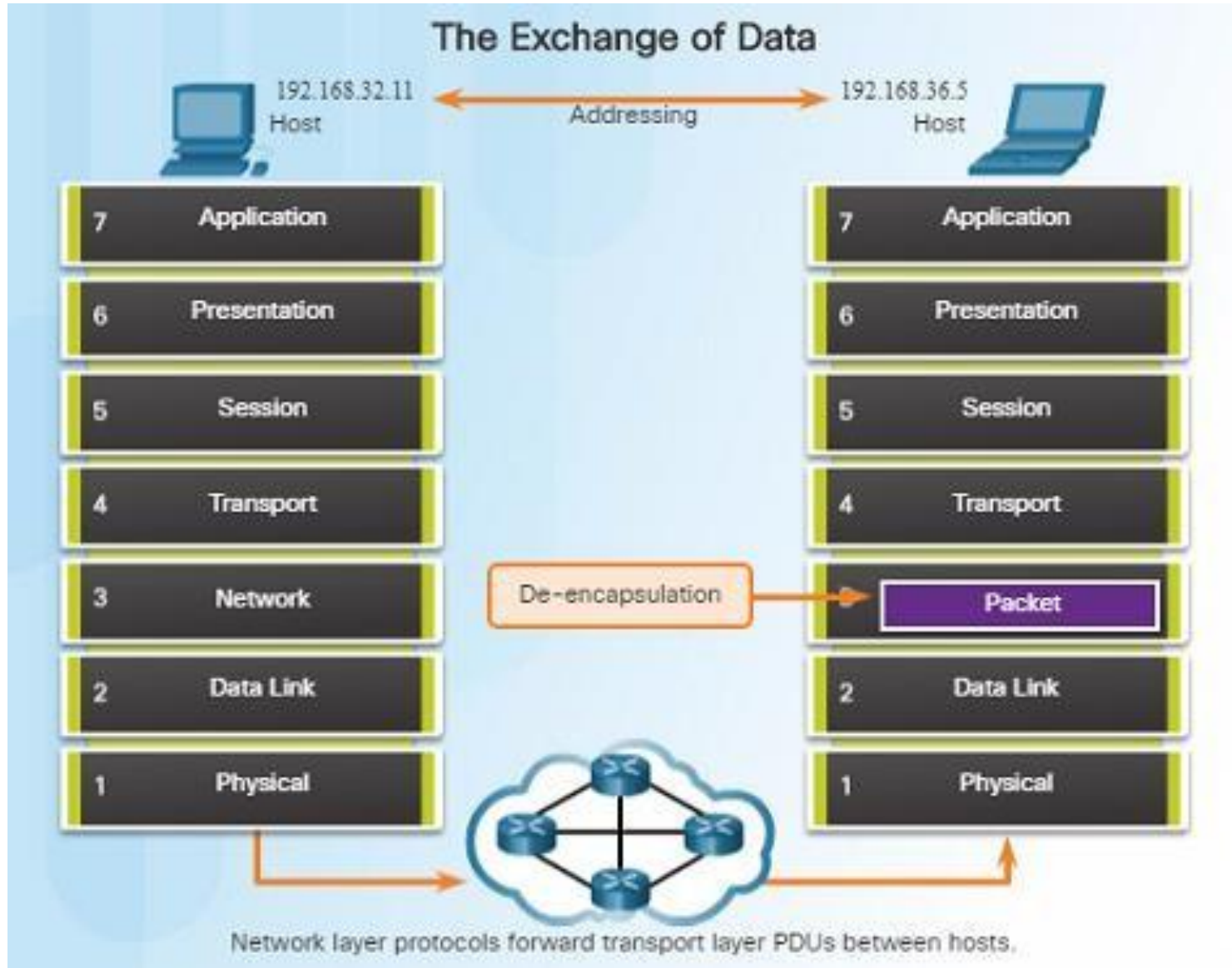
The Network Layer

- The network layer, which resides at OSI Layer 3, provides services that **allow end devices to exchange data across a network.**
- The network layer uses **four processes** in order to provide end-to-end transport:
 - **Addressing of end devices** – IP addresses must be unique for identification purposes.
 - **Encapsulation** – The protocol data units from the transport layer are encapsulated by adding IP header information including source and destination IP addresses.
 - **Routing** – The network layer provides services to direct packets to other networks. Routers select the best path for a packet to take to its destination network.
 - **De-encapsulation** – The destination host de-encapsulates the packet to see if it matches its own.



Network Layer in Communications

The Network Layer

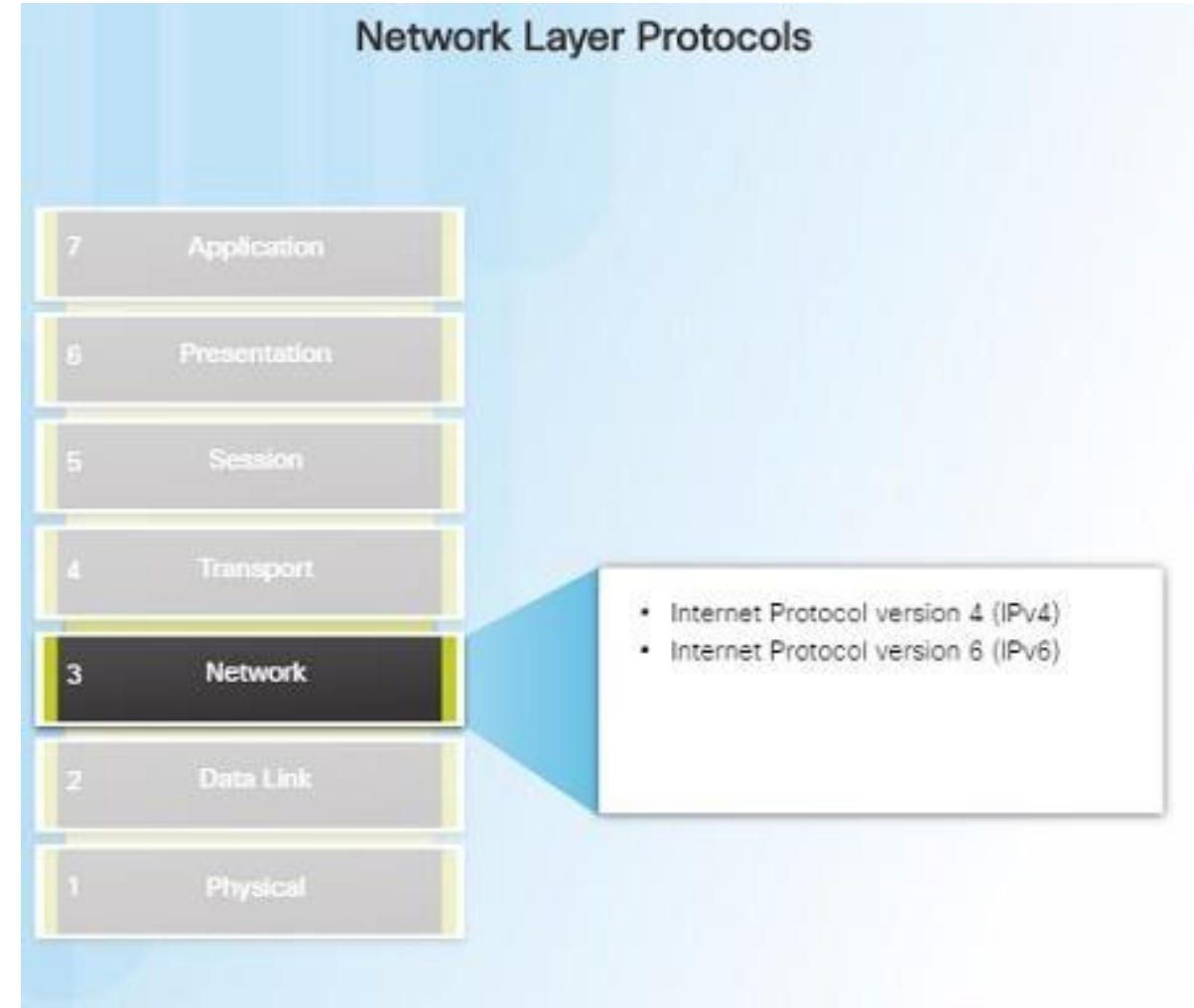




Network Layer in Communications

Network Layer Protocols

- There are several network layer protocols in existence; however, the most commonly implemented are:
 - **Internet Protocol version 4 (IPv4)**
 - **Internet Protocol version 6 (IPv6)**
- Note: Legacy network layer protocols are not discussed in this course.
 - Novell Internetwork Packet Exchange (IPX)
 - AppleTalk
 - Connectionless Network Service (CLNS/DECNet)





Characteristics of the IP Protocol

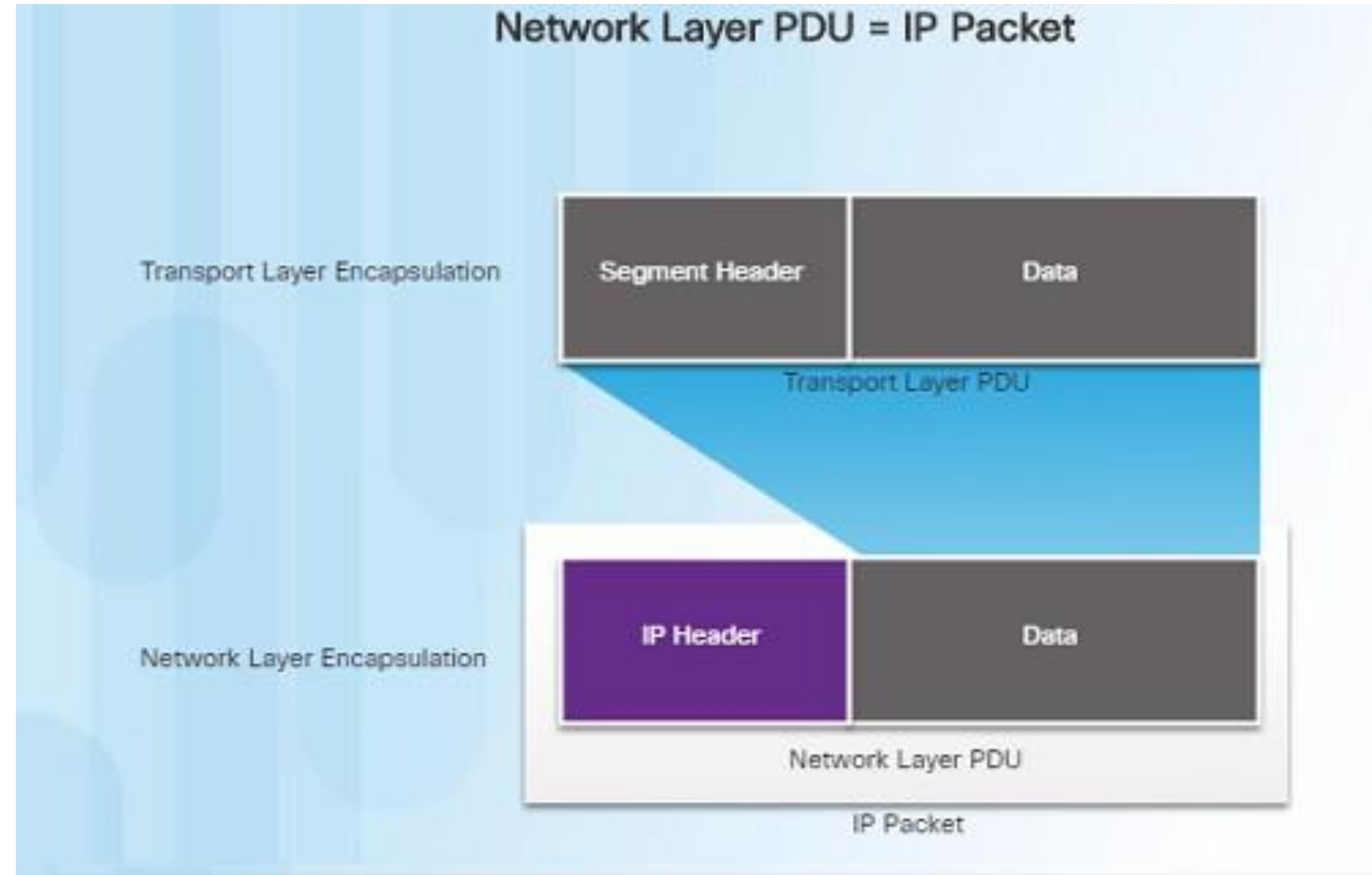
Encapsulating IP

- At the network layer, IP encapsulates the transport layer segment by adding an IP header for the purpose of delivery to the destination host.
- The **IP header stays the same from the source to the destination host.**
- The process of encapsulating data layer by layer enables the services at different layers to scale without affecting other layers.
- Routers implement different network layer protocols concurrently over a network **and use the network layer packet header for routing.**



Characteristics of the IP Protocol

Encapsulating IP



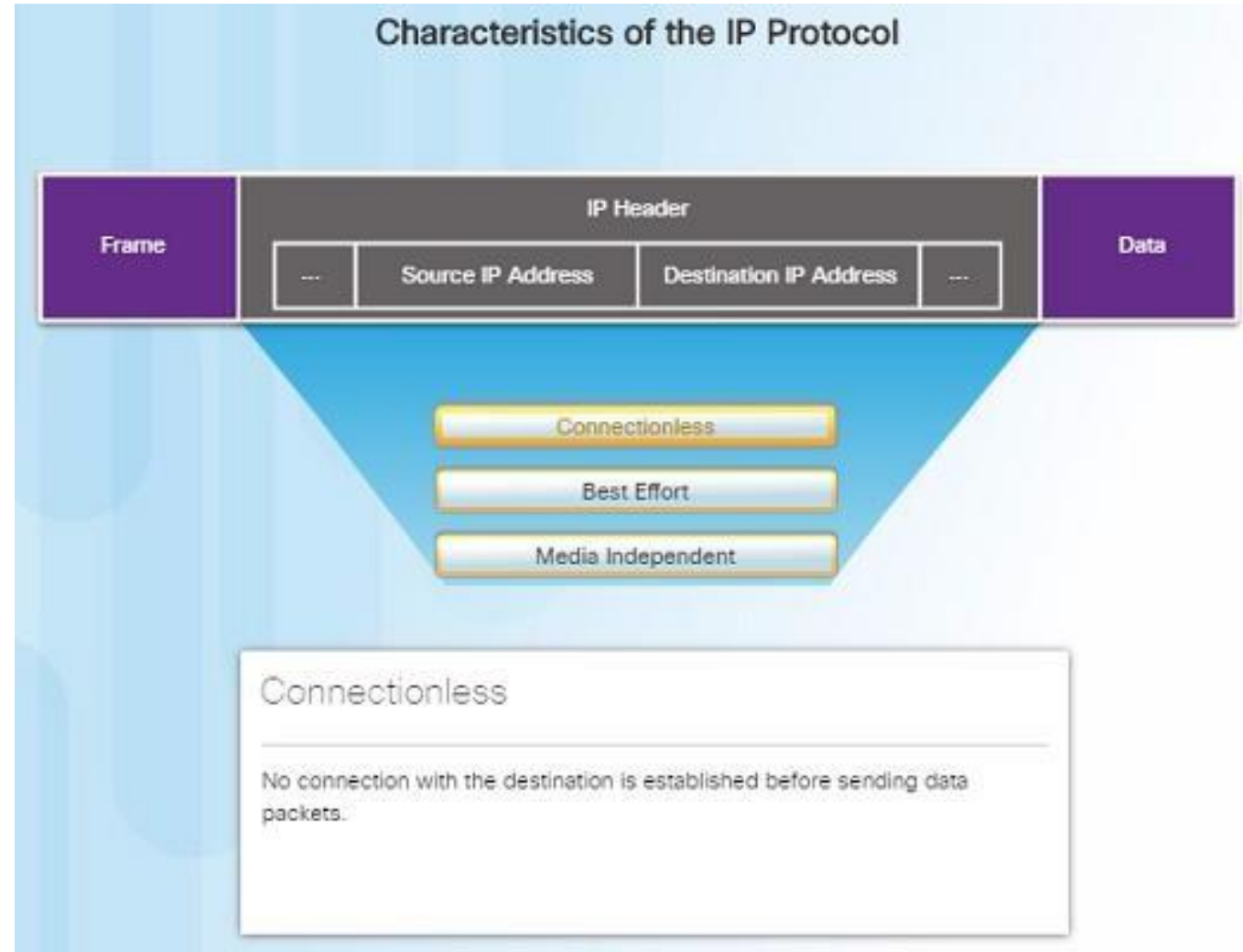
The transport layer adds a header so segments can be reassembled at the destination.

The network layer adds a header so packets can be routed through complex networks and reach their destination. In TCP/IP based networks, the network layer PDU is the IP Packet.

Characteristics of the IP Protocol

Characteristics of IP

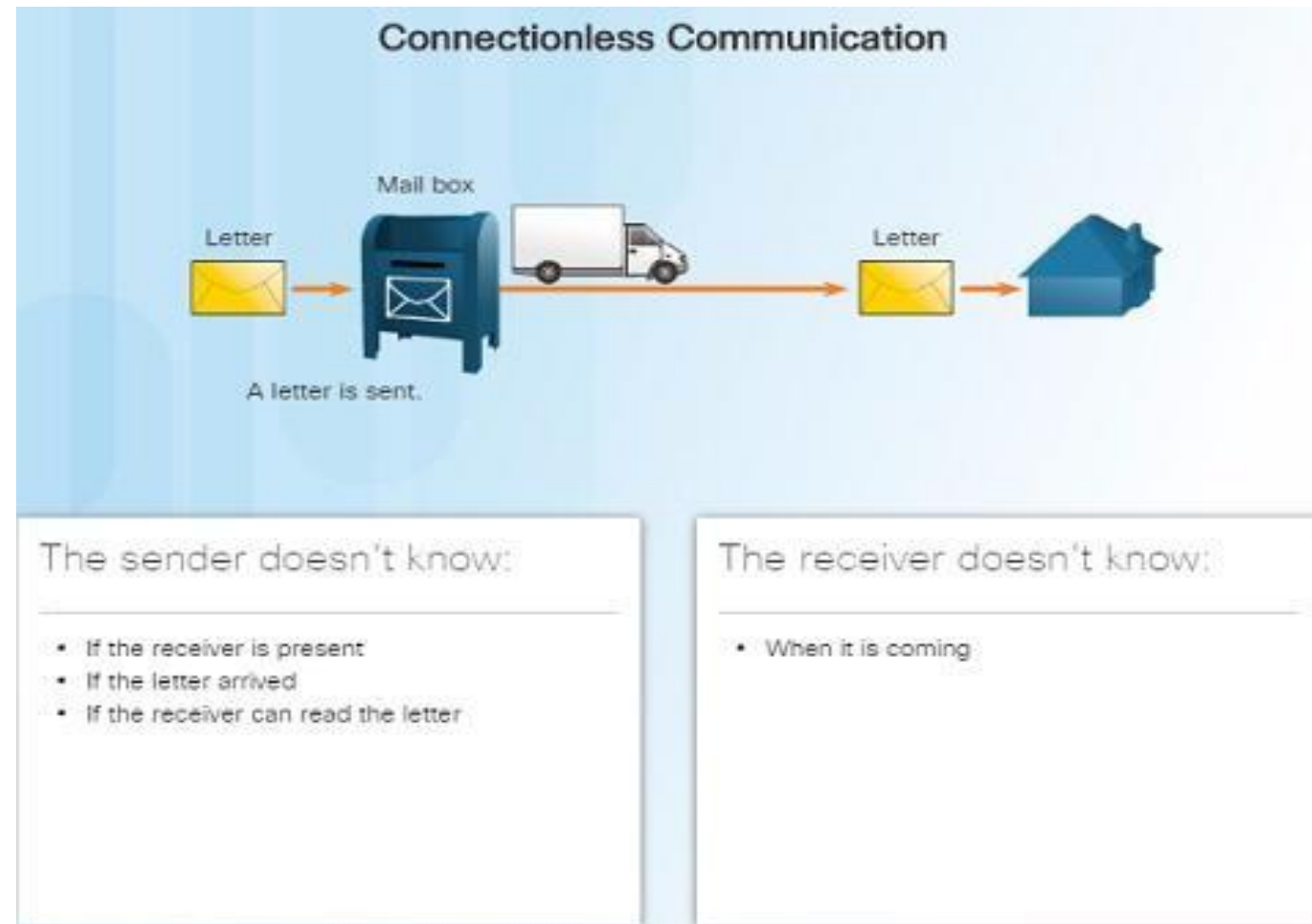
- IP was designed as a protocol with **low overhead** – it provides only the functions required to deliver a packet from the source to a destination.
- An IP packet is sent to the destination **without prior establishment of a connection**
- IP was **not designed to track and manage the flow of** packets.
- These functions, if required, are performed by other layers – primarily TCP



Characteristics of the IP Protocol

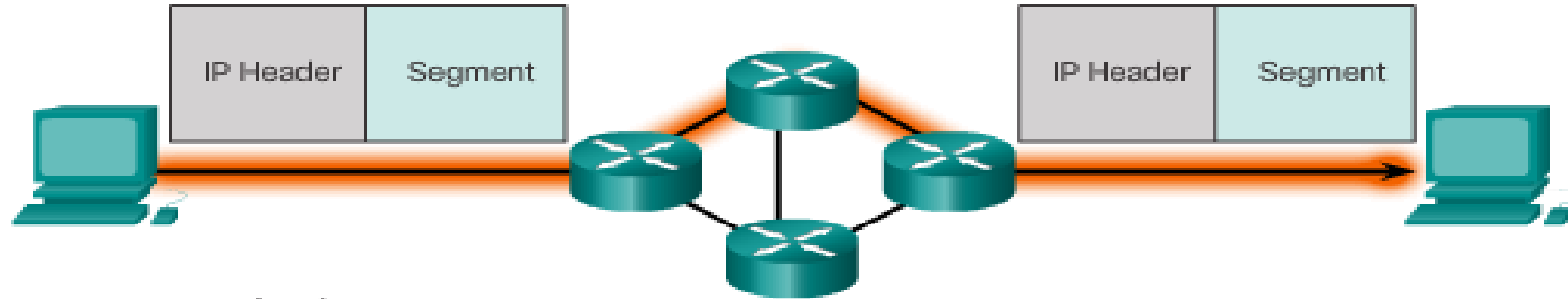
IP - Connectionless

- IP is a connectionless protocol:
- **No dedicated end-to-end connection** is created before data is sent.
- Very similar process as sending someone a letter through snail mail.
- Senders do not know whether or not the destination is present, reachable, or functional before sending packets.
- **This feature contributes to the low overhead of IP.**





IP – Connectionless (cont.)



A packet is sent.

The sender doesn't know:

- If the receiver is present
- If the packet arrived
- If the receiver can read the packet

The receiver doesn't know:

- When it is coming

out-of-order or missing packets



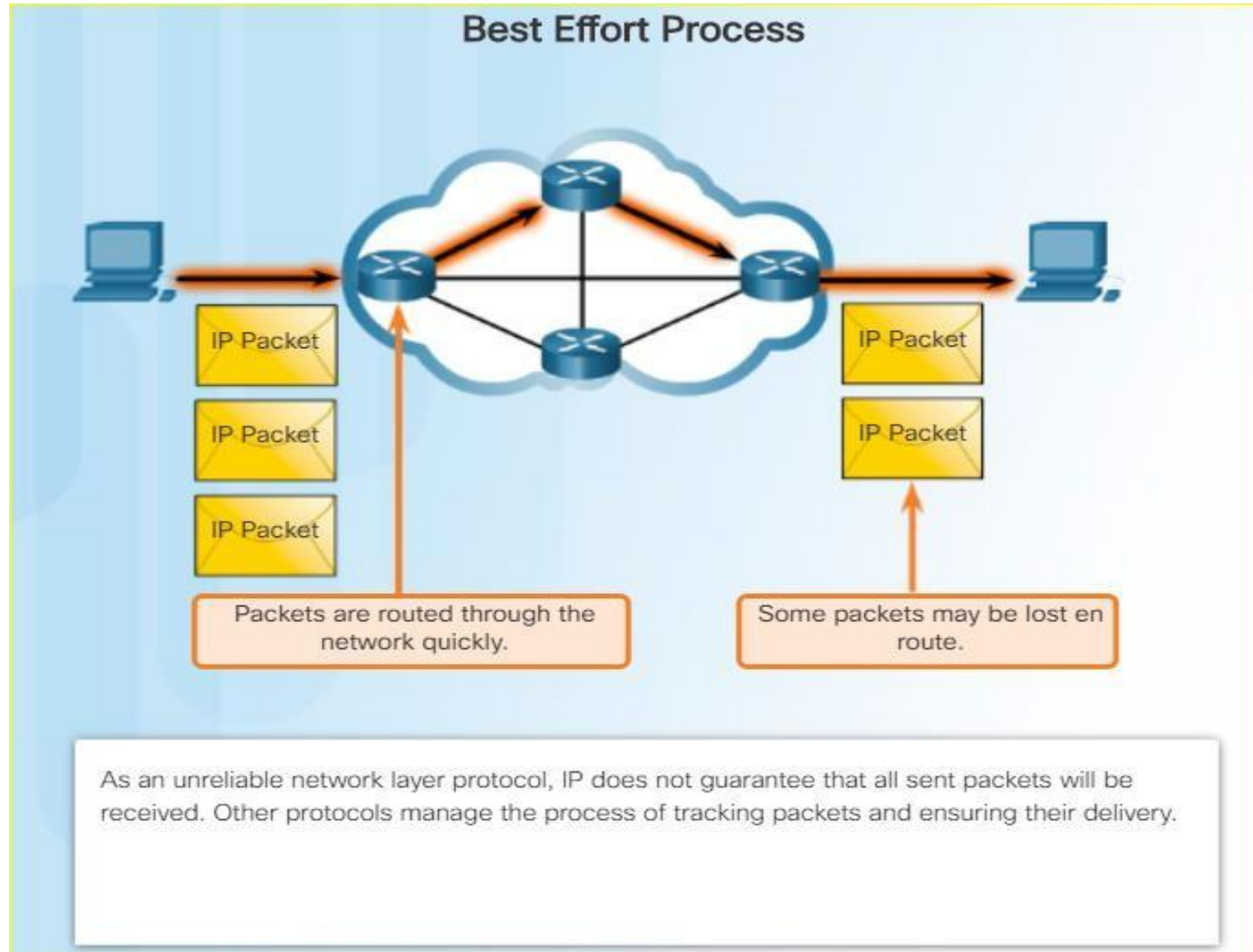
Characteristics of the IP Protocol

IP – Best Effort Delivery

- IP is a Best Effort Delivery protocol:
 - IP is considered “unreliable” because it does not guarantee that all packets that are sent will be received.
 - Unreliable means that IP does not have the capability to manage and recover from undelivered, corrupt, or out of sequence packets.
 - If packets are missing or not in the correct order at the destination, upper layer protocols/services must resolve these issues.

Characteristics of the IP Protocol

IP – Best Effort Delivery





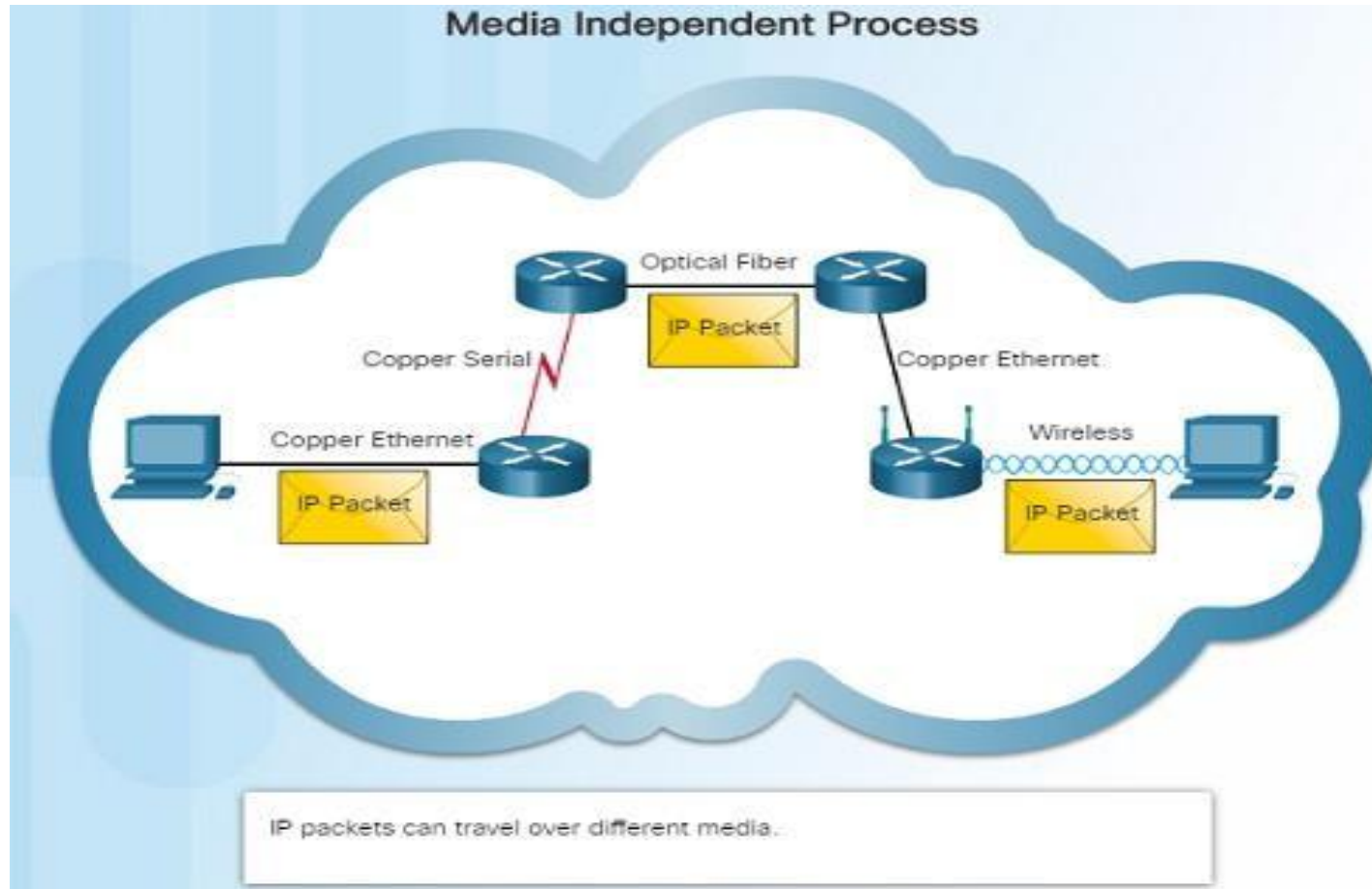
Characteristics of the IP Protocol

IP – Media Independent

- **IP operates independently from the media** that carries the data at lower layers of the protocol stack – it does not care if the media is copper cables, fiber optics or wireless.
 - The OSI data link layer is responsible for taking the IP packet and preparing it for transmission over the communications medium.
 - The network layer does have a **maximum size of the PDU** that can be transported – referred to as **MTU (maximum transmission unit)**.
 - The data link layer tells the network layer the MTU.
-
- **Fragmentation:** At times, an intermediary device (router) will need to split up a packet when forwarding it from one media to a media with a smaller MTU.

Characteristics of the IP Protocol

IP – Media Independent



- **Copper Ethernet:** MTU = 1,518 bytes.
- **Copper Serial:** Frame Relay MTU = 512 bytes.
- **Optical Fiber:** ATM MTU = 17,966 bytes.
- **Wireless:** 802.11 MTU = 2272 bytes.

IPv4 Packet

Module 8: Network Later

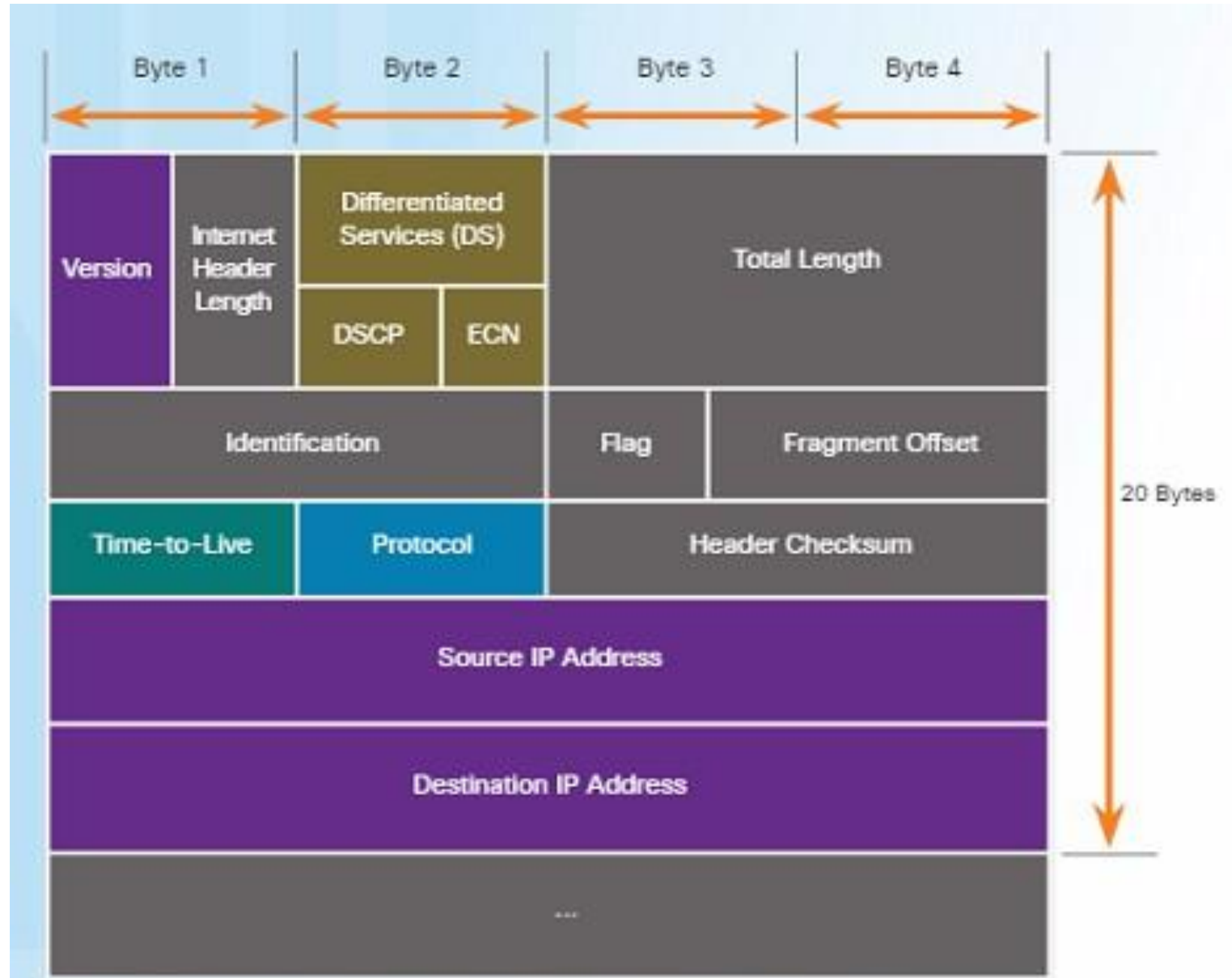
Introduction to Networks v7.0 (ITN)



IPv4 Packet

IPv4 Packet Header

- The IPv4 network header characteristics:
 - It is in binary.
 - Contains several fields of information
 - Diagram is read from left to right, 4 bytes per line
 - The two most important fields are the source and destination.



Protocols may have may have one or more functions.



IPv4 Packet

IPv4 Packet Header

- An IPv4 packet header consists of the fields containing binary numbers. These numbers identify various settings of the IP packet which are examined by the Layer 3 process.
- Significant fields include:
 - **Version** – Specifies that the packet is IP version 4
 - **Differentiated Services or DiffServ (DS)** – Used to determine the **priority** of each packet on the network.
 - **Time-to-Live (TTL)** – **Limits the lifetime of a packet** – decreased by one at each router along the way.
 - **Protocol** – Used to identify the **next level protocol**.
 - **Source IPv4 Address** – Source address of the packet.
 - **Destination IPv4 Address** – Address of destination.

Video Demonstration – Sample IPv4 Headers in Wireshark

- Wireshark is a free and open source packet and network protocol analyzer that allows you to capture and browse network traffic.



IPv6 Packets

Module 8: Network Layer

Limitations of IPv4

- IPv4 has been updated to address new challenges.
- Three major issues still exist with IPv4:
 - **IP address depletion** – IPv4 has a limited number of unique public IPv4 addresses available. Although there are about 4 billion IPv4 addresses, the exponential growth of new IP-enabled devices has increased the need.
 - **Internet routing table expansion** – A routing table contains the routes to different networks in order to make the best path determination. As more devices and servers are connected to the network, more routes are created. A large number of routes can slow down a router.
 - **Lack of end-to-end connectivity** – Network Address Translation (NAT) was created for devices to share a single IPv4 address. However, because they are shared, this can cause problems for technologies that require end-to-end connectivity.





IPv6 Packet

Introducing IPv6

- In the early '90s, the IETF started looking at a replacement for IPv4 – which led to IPv6.
- Advantages of IPv6 over IPv4 include:
 - **Increased address space** – based on 128-bit addressing vs. 32-bit with IPv4
 - **Improved packet handling** – fewer fields with IPv6 than IPv4
 - **Eliminates the need for NAT** – no need to share addresses with IPv6

- There are roughly enough IPv6 addresses for every grain of sand on Earth.

How Many Addresses Are Available with IPv6?

Number Name	Scientific Notation	Number of Zeros
1 Thousand	10^3	1,000
1 Million	10^6	1,000,000
1 Billion	10^9	1,000,000,000
1 Trillion	10^{12}	1,000,000,000,000
1 Quadrillion	10^{15}	1,000,000,000,000,000
1 Quintillion	10^{18}	1,000,000,000,000,000,000
1 Sextillion	10^{21}	1,000,000,000,000,000,000,000
1 Septillion	10^{24}	1,000,000,000,000,000,000,000,000
1 Octillion	10^{27}	1,000,000,000,000,000,000,000,000,000
1 Nonillion	10^{30}	1,000,000,000,000,000,000,000,000,000,000
1 Decillion	10^{33}	1,000,000,000,000,000,000,000,000,000,000,000
1 Undecillion	10^{36}	1,000,000,000,000,000,000,000,000,000,000,000,000

Legend

- There are 4 billion IPv4 addresses
- There are 340 undecillion IPv6 addresses



IPv6 Packet

Encapsulating IPv6

The IPv6 header is simpler than the IPv4 header.

IPv4 and IPv6 Headers

IPv4 Header

Version	IHL	Type of Service	Total Length	
Identification			Flags	Fragment Offset
Time to Live	Protocol		Header Checksum	
Source Address				
Destination Address				
Options			Padding	

IPv6 Header

Version	Traffic Class	Flow Label	
Payload Length		Next Header	Hop Limit
Source Address			
Destination Address			

Legend

- Field names kept from IPv4 to IPv6
- Fields not kept in IPv6
- Name & position changed in IPv6
- New field in IPv6



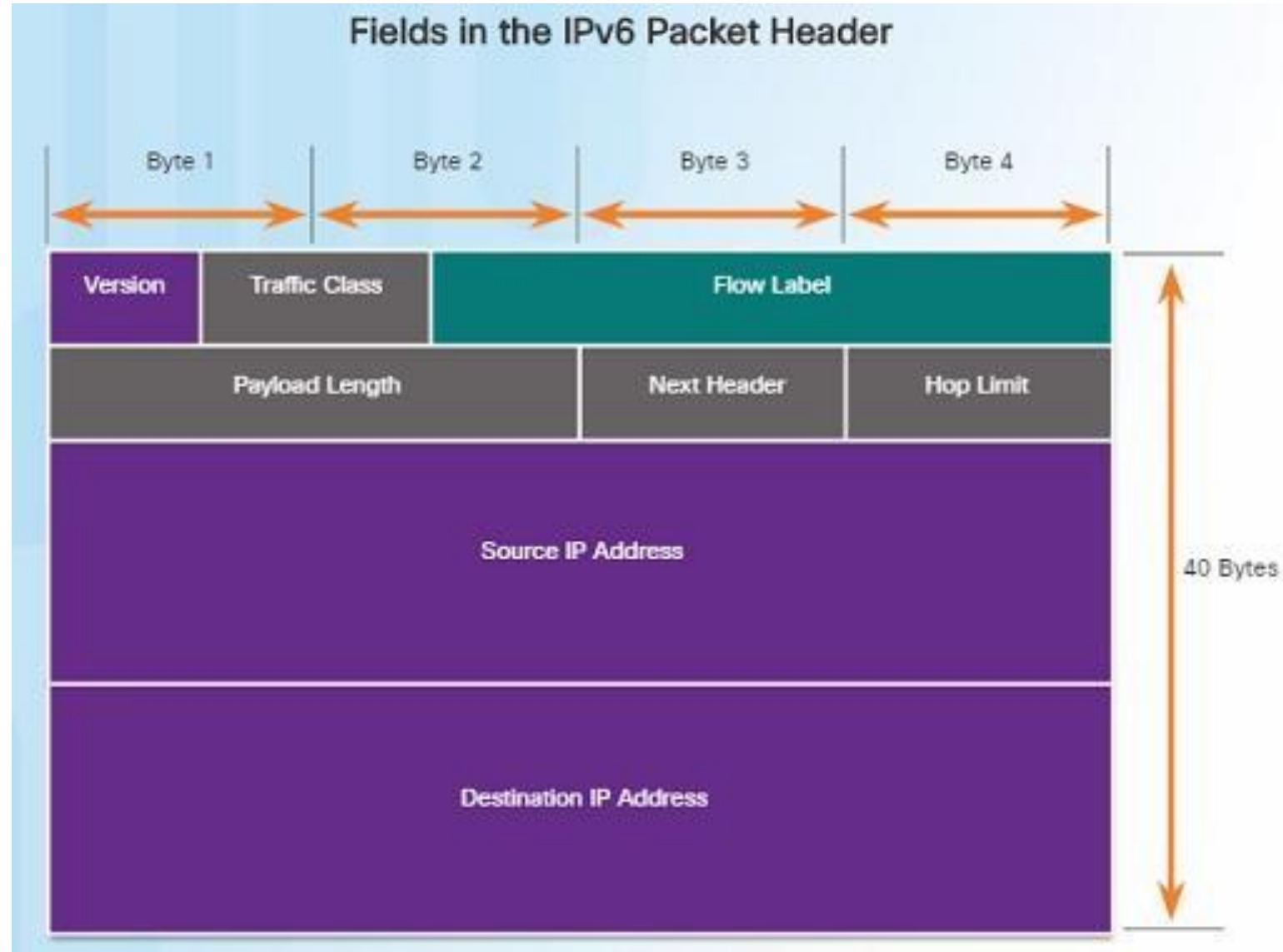
Encapsulating IPv6 (Cont.)

- Advantages of IPv6 over IPv4 using the simplified header:
 - **Simplified header format** for efficient packet handling
 - Hierarchical network architecture for **routing efficiency**
 - **Autoconfiguration** for addresses
 - **Elimination of need for network address translation (NAT) between private and public addresses**



IPv6 Packet

IPv6 Packet Header





IPv6 Packet Header

- IPv6 packet header fields:
 - Version – Contains a 4-bit binary value set to 0110 that identifies it as a IPv6 packet.
 - Traffic Class – 8-bit field equivalent to the IPv4 Differentiated Services (DS) field.
 - Flow Label – 20-bit field suggests that all packets with the same flow label receive the same type of handling by routers.
 - Payload Length – 16-bit field indicates the length of the data portion or payload of the packet.
 - Next Header – 8-bit field is equivalent to the IPv4 Protocol field. It indicates the data payload type that the packet is carrying.



IPv6 Packet Header (Cont.)

- IPv6 packet header fields:
 - **Hop Limit** – 8-bit field replaces the IPv4 TTL field. This value is decremented by 1 as it passes through each router. When **it reaches zero, the packet is discarded**.
 - **Source IPv6 Address** – **128-bit** field that identifies the IPv6 address of the sending host.
 - **Destination IPv6 Address** – **128-bit** field that identifies the IPv6 address of the receiving host.



IPv6 Packets

IPv6 Packet Header (Cont.)

IPv6 packet may also contain extension headers (EH).

EH headers characteristics:

- provide optional network layer information
- are optional
- are placed between IPv6 header and the payload
- may be used for fragmentation, security, mobility support, etc.

Note: Unlike IPv4, routers do not fragment IPv6 packets.



IPv6 Packet

Sample IPv6 Header – Video



Demonstration

Sample IPv6 Headers in Wireshark

8.2 Routing

8.4 How a Host Routes

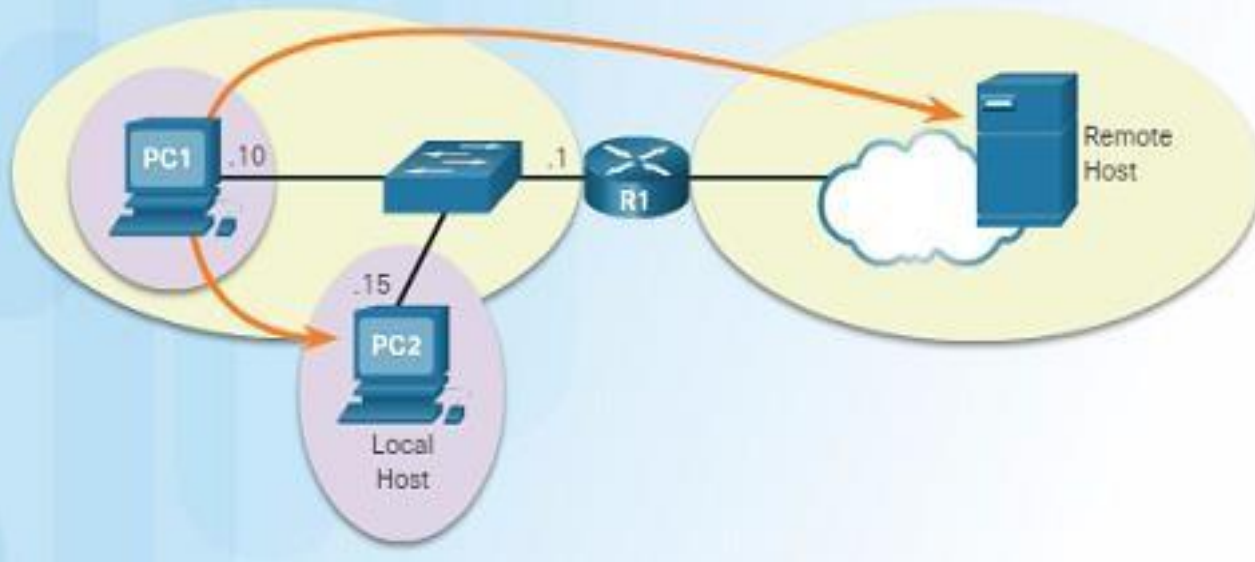
Module 8: Network Later

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How a Host Routes

Host Forwarding Decision

Three Types of Destinations



- An important role of the network layer is to direct packets between hosts. A host can send a packet to:
 - **Itself** – A host can ping itself for testing purposes using 127.0.0.1 which is referred to as the loopback interface.
 - **Local host** – This is a host on the same local network as the sending host. The hosts share the same network address.
 - **Remote host** – This is a host on a remote network. The hosts do not share the same network address.
- The source IPv4 address and subnet mask is compared with the destination address and subnet mask in order to determine if the host is on the local network or remote network.

How a Host Routes Default Gateway

Default Gateway Functions

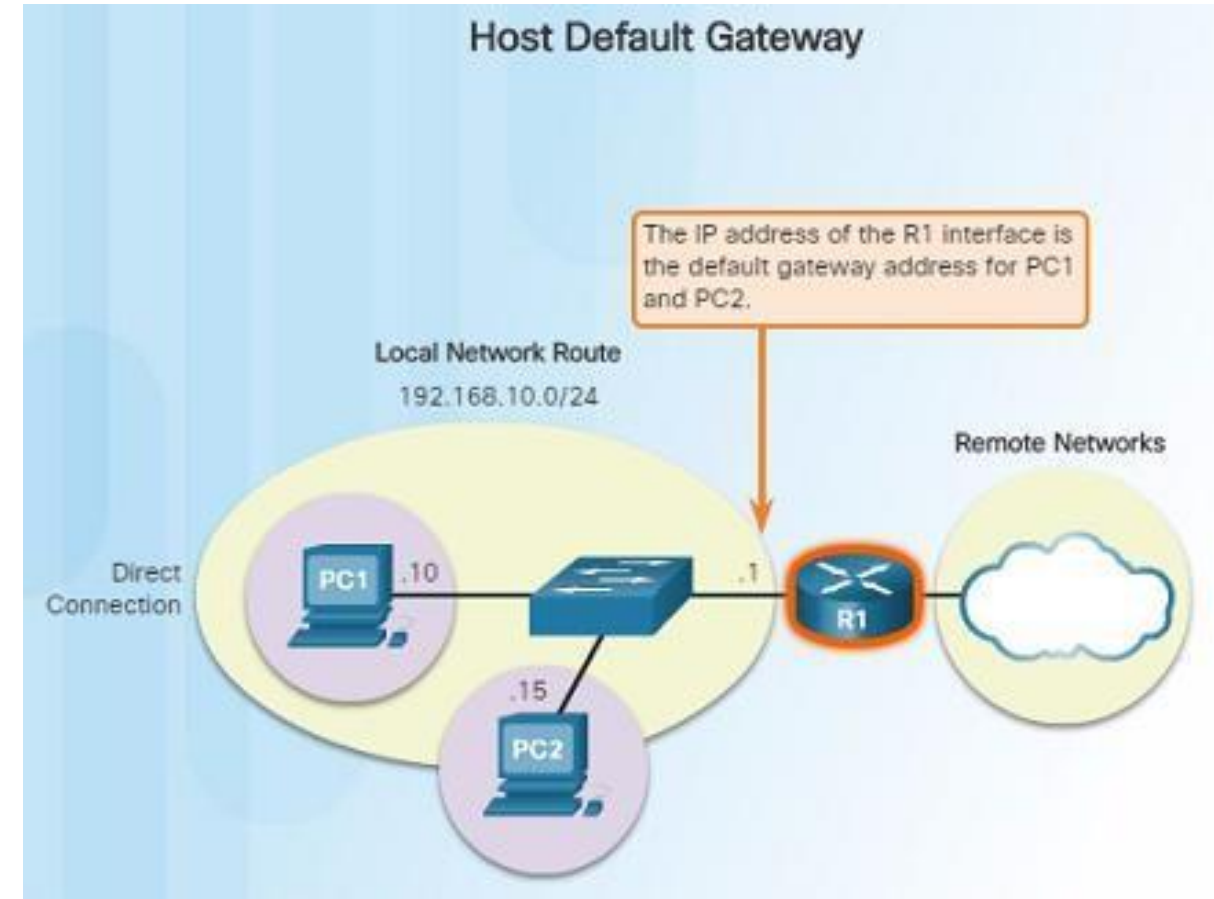
A Default Gateway ...

- Routes traffic to other networks.
- Has a local IP address in the same address range as other hosts on the network.
- Can take data in and forward data out.

- The default gateway is the network device that can route traffic out to other networks. It is the **router** that routes traffic out of a local network.
- This occurs when the destination host is not on the same local network as the sending host.
- The default gateway will know where to send the packet using its **routing table**.
- The sending host does not need to know where to send the packet other than to the default gateway – or router.

How a Host Routes Using the Default Gateway

- A host's routing table usually includes a default gateway address – which is the router IP address for the network that the host is on.
- The host receives the IPv4 address for the default gateway from DHCP, or it is manually configured.
- Having a default gateway configured creates a default route in the routing table of a host - which is the route the computer will send a packet to when it needs to contact a remote network.



How a Host Routes

Host Routing Tables



IPv4 Routing Table for PC1

```
C:\Users\PC1> netstat -r
```

IPv4 Route Table

Active Routes:

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	192.168.10.1	192.168.10.10	25
127.0.0.0	255.0.0.0	On-link	127.0.0.1	306
127.0.0.1	255.255.255.255	On-link	127.0.0.1	306
127.255.255.255	255.255.255.255	On-link	127.0.0.1	306
192.168.10.0	255.255.255.0	On-link	192.168.10.10	281
192.168.10.10	255.255.255.255	On-link	192.168.10.10	281
192.168.10.255	255.255.255.255	On-link	192.168.10.10	281
224.0.0.0	240.0.0.0	On-link	127.0.0.1	306
224.0.0.0	240.0.0.0	On-link	192.168.10.10	281
255.255.255.255	255.255.255.255	On-link	127.0.0.1	306
255.255.255.255	255.255.255.255	On-link	192.168.10.10	281

- On a Windows host, you can display the routing table using:
 - **route print**
 - **netstat -r**
- Three sections will be displayed:
 - Interface List – Lists the Media Access Control (MAC) address and assigned interface number of network interfaces on the host.
 - IPv4 Route Table – Lists all known IPv4 routes.
 - IPv6 Route Table – Lists all known IPv6 routes.

Introduction to Routing

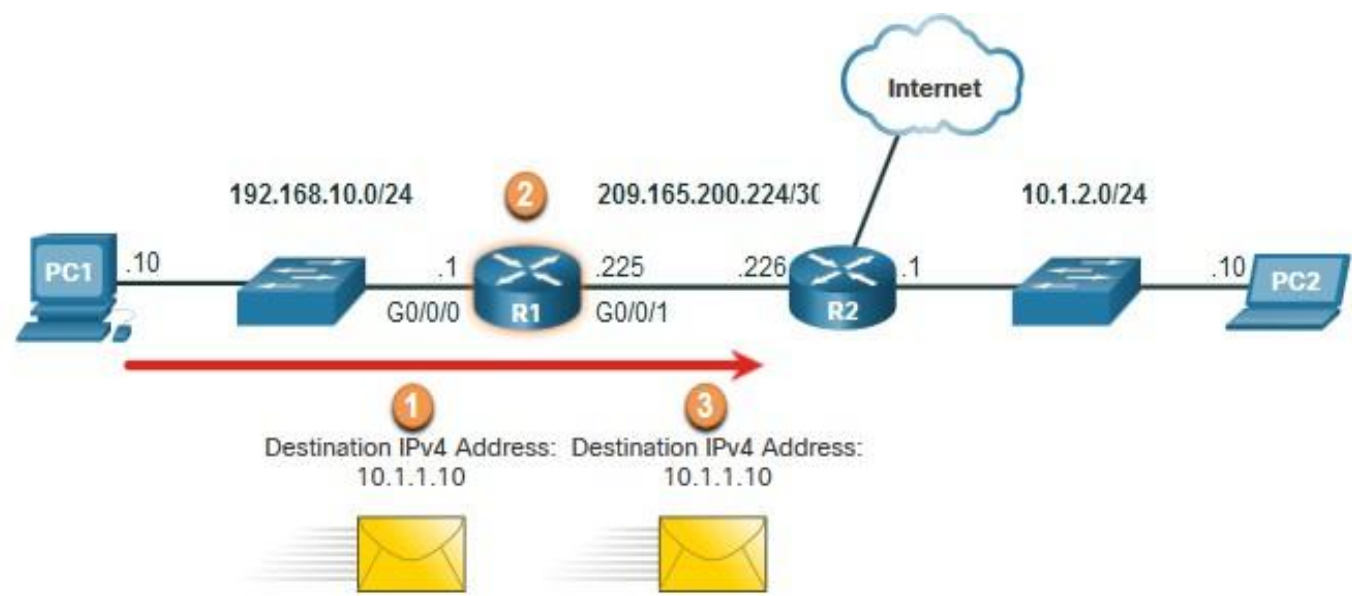
Module 8: Network Layer

Introduction to Networks v7.0 (ITN)

Introduction to Routing

Router Packet Forwarding Decision

What happens when the router receives the frame from the host device?



R1 Routing Table

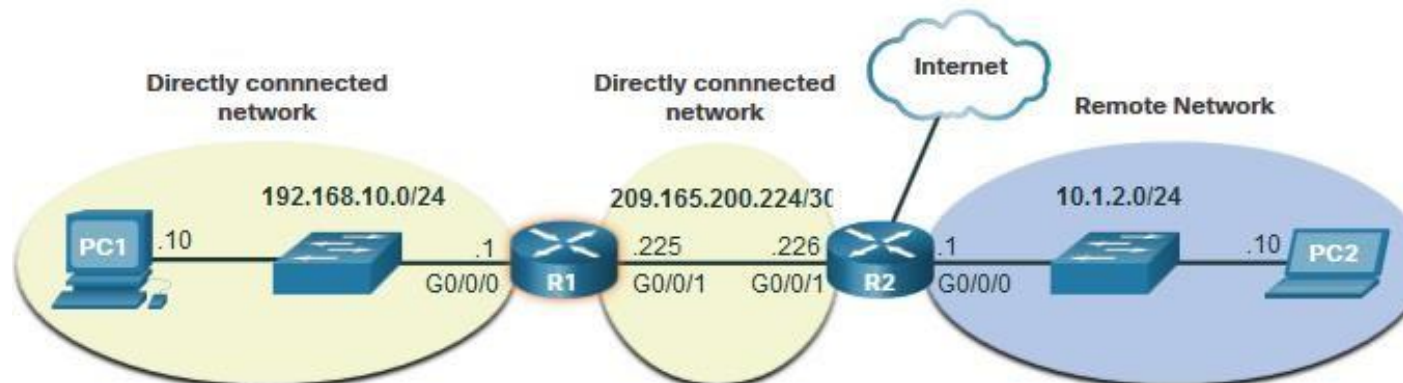
Route	Next Hop or Exit Interface
192.168.10.0 /24	G0/0/0
209.165.200.224/30	G0/0/1
10.1.1.0/24	via R2
Default Route 0.0.0.0/0	via R2

1. Packet arrives on the Gigabit Ethernet 0/0/0 interface of router R1. R1 de-encapsulates the Layer 2 Ethernet header and trailer.
2. Router R1 examines the destination IPv4 address of the packet and searches for the best match in its IPv4 routing table. The route entry indicates that this packet is to be forwarded to router R2.
3. Router R1 encapsulates the packet into a new Ethernet header and trailer, and forwards the packet to the next hop router R2.

IP Router Routing Table

There are three types of routes in a router's routing table:

- **Directly Connected** – These routes are automatically added by the router, provided the interface is active and has addressing.
- **Remote** – These are the routes the router does not have a direct connection and may be learned:
 - Manually – with a static route
 - Dynamically – by using a routing protocol to have the routers share their information with each other
- **Default Route** – this forwards all traffic to a specific direction when there is not a match in the routing table

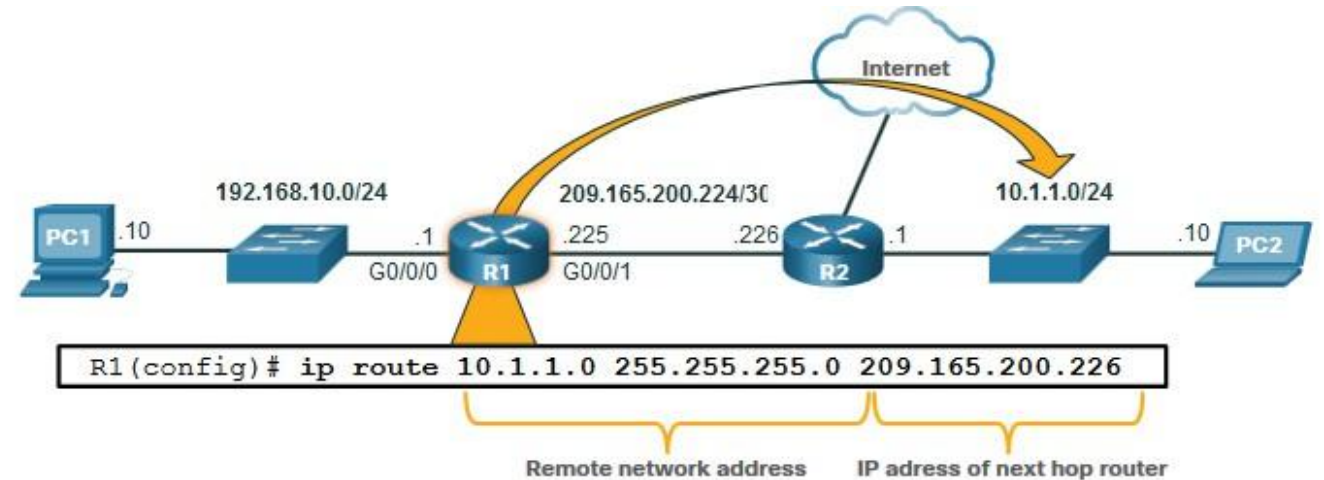


Introduction to Routing

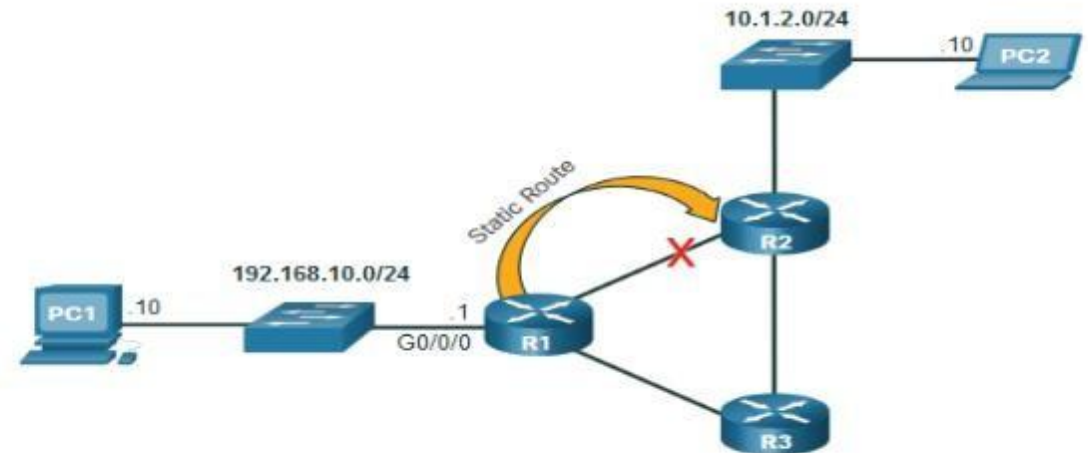
Static Routing

Static Route Characteristics:

- Must be configured manually
- Must be adjusted manually by the administrator when there is a change in the topology
- Good for small non-redundant networks
- Often used in conjunction with a dynamic routing protocol for configuring a default route



R1 is manually configured with a static route to reach the 10.1.1.0/24 network. If this path changes, R1 will require a new static route.



If the route from R1 via R2 is no longer available, a new static route via R3 would need to be configured. A static route does not automatically adjust for topology changes.

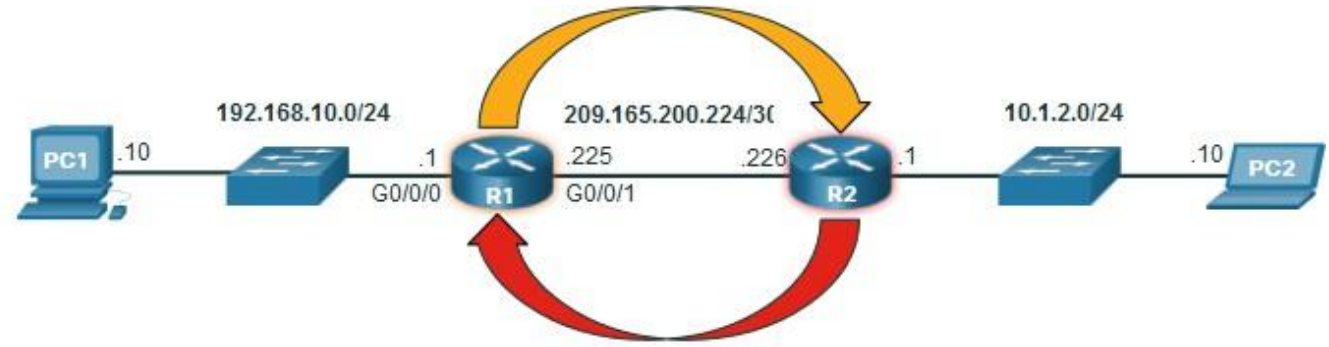
Introduction to Routing

Dynamic Routing

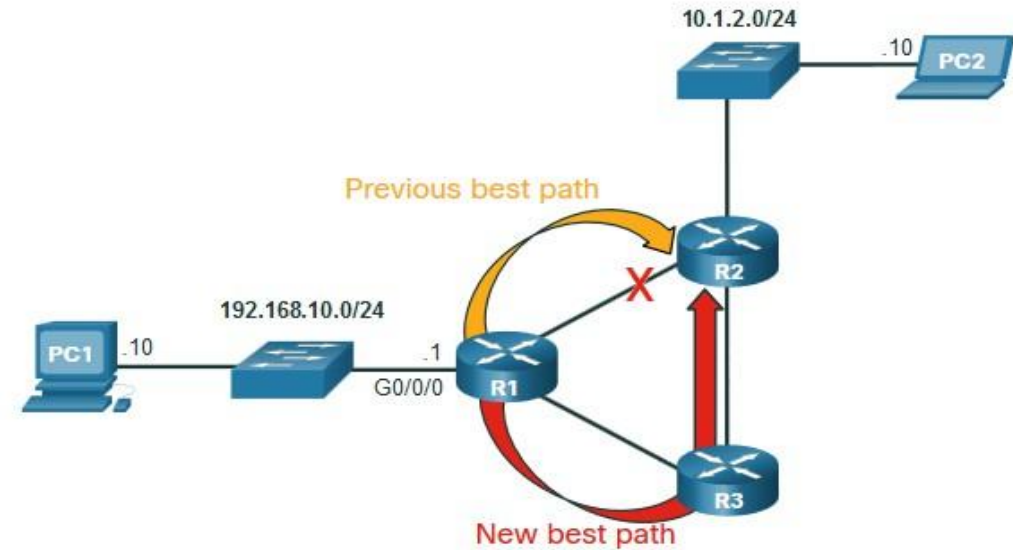
Dynamic Routes Automatically:

- Discover remote networks
- Maintain up-to-date information
- Choose the best path to the destination
- Find new best paths when there is a topology change

Dynamic routing can also share static default routes with the other routers.



- R1 is using the routing protocol OSPF to let R2 know about the 192.168.10.0/24 network.
- R2 is using the routing protocol OSPF to let R1 know about the 10.1.1.0/24 network.



R1, R2, and R3 are using the dynamic routing protocol OSPF. If there is a network topology change, they can automatically adjust to find a new best path.

Explaining the IPv4 Routing Table – Video



IPv4 Router Routing Table

- The **show ip route** command is used to display the router's IPv4 routing table. The routing table shows:
 - Directly connected and remote routes
 - How each route was learned
 - Trustworthiness and rating of the route
 - When the route was last updated
 - Which interface is used to reach the destination
- A router examines an incoming packet's header to determine the destination network. If there's a match, the packet is forwarded using the specified information in the routing table.

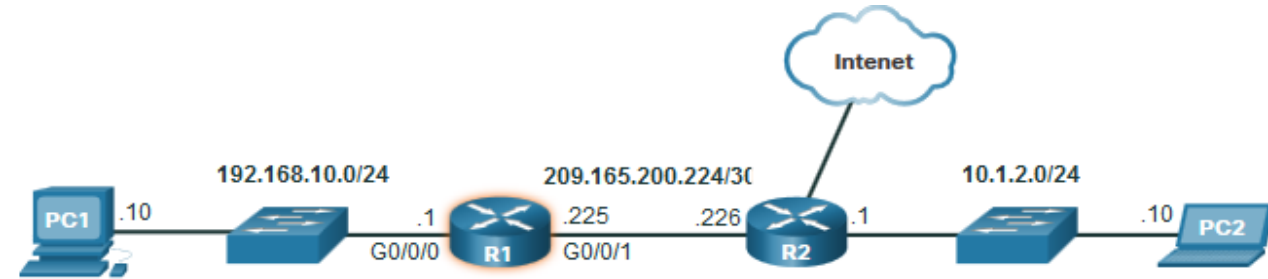
Introduction to an IPv4 Routing Table

The **show ip route** command shows the following route sources:

- **L** - Directly connected local interface IP address
- **C** – Directly connected network
- **S** – Static route was manually configured by an administrator
- **O** – OSPF
- **D** – EIGRP

This command shows types of routes:

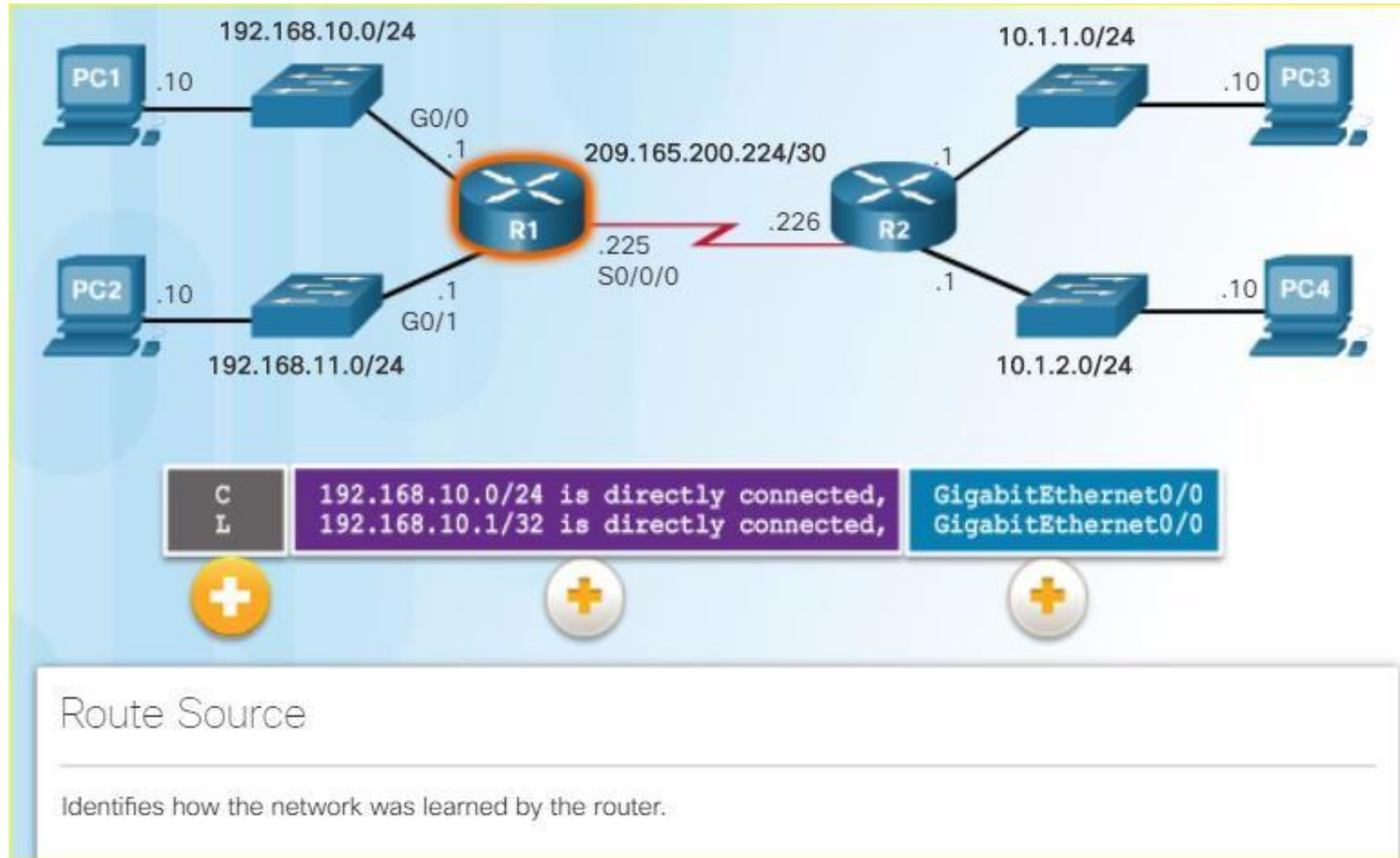
- Directly Connected – C and L
- Remote Routes – O, D, etc.
- Default Routes – S*



```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 209.165.200.226 to network 0.0.0.0
S*   0.0.0.0/0 [1/0] via 209.165.200.226, GigabitEthernet0/0/1
     10.0.0.0/24 is subnetted, 1 subnets
O     10.1.1.0 [110/2] via 209.165.200.226, 00:02:45, GigabitEthernet0/0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L     192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
     209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C     209.165.200.224/30 is directly connected, GigabitEthernet0/0/1
L     209.165.200.225/32 is directly connected, GigabitEthernet0/0/1
R1#
```

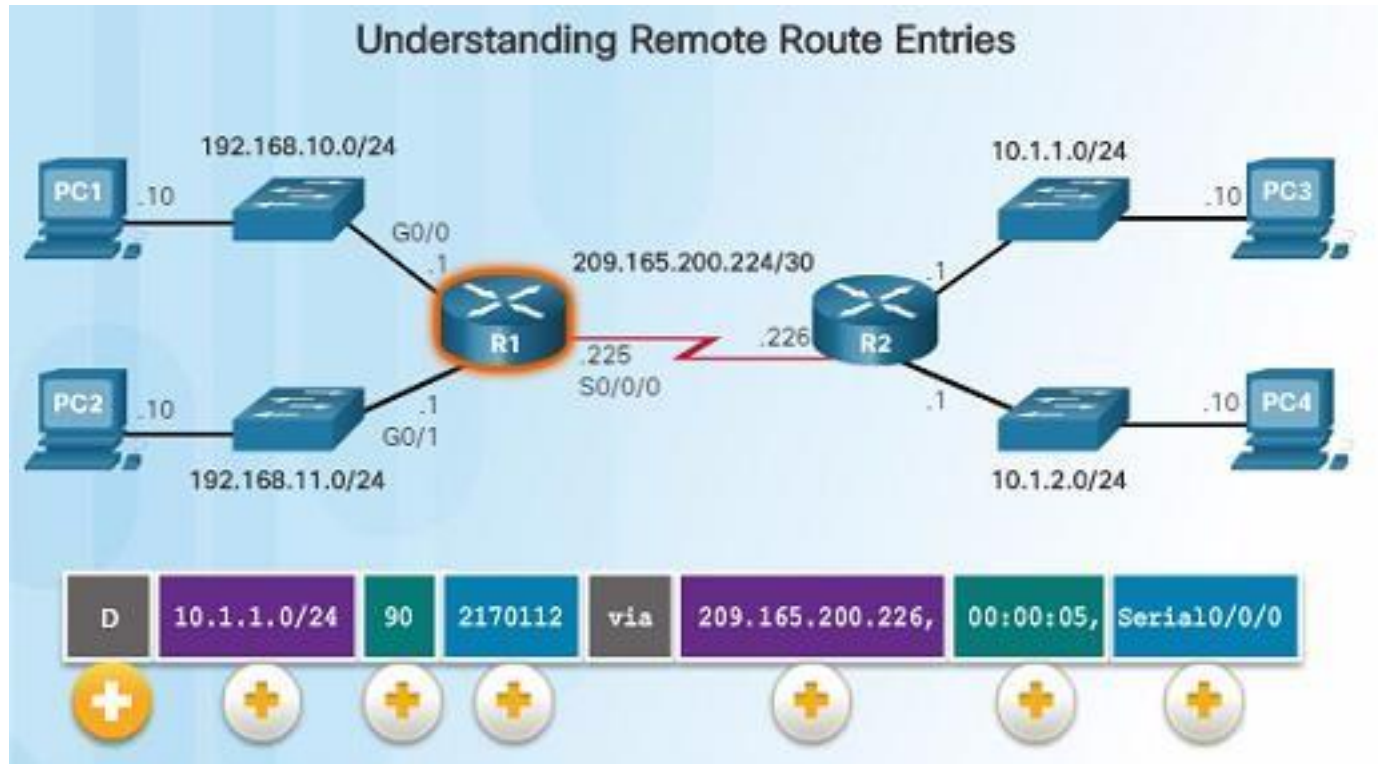

Directly Connected Routing Table Entries



- When a router interface is configured and activated, the following two routing table entries are created automatically:
 - **C** – Identifies that the **network is directly connected** and the interface is configured with an IP address and activated.
 - **L** – Identifies that it is a **local interface**. This is the IPv4 address of the interface on the router.

Router Routing Tables

Understanding Remote Route Entries

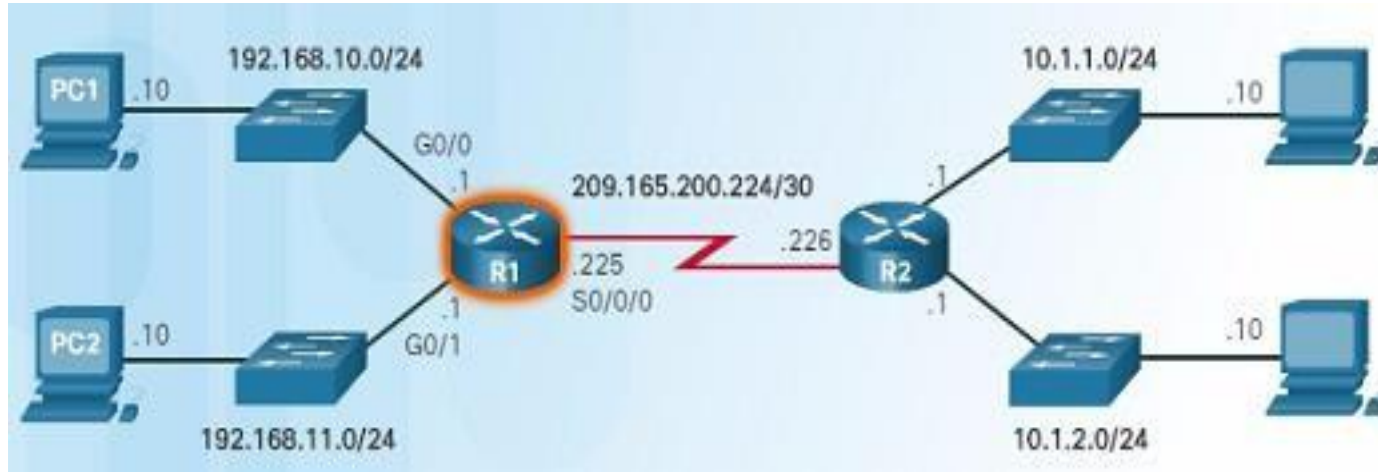


- The **D** represents the Route Source which is how the network was learned by the router. **D** identifies the route as an **EIGRP route** or (Enhanced Interior Gateway Routing Protocol)

- 10.1.1.0/24** identifies the destination network.
- 90** is the **administrative distance** for the corresponding network – or the trustworthiness of the route. **The lower the number, the more trustworthy it is.**
- 2170112** – represents the **metric** or value assigned to reach the remote network. **Lower values indicate preferred routes.**
- 209.165.200.226** – **Next-hop** or IP address of the next router to forward the packet.
- 00:00:05** - Route Timestamp identifies when the router was last heard from.
- Serial/0/0/0** – **Outgoing Interface**

Router Routing Tables

Next-Hop Address



```
R1# show ip route
<output omitted>
Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
D   10.1.1.0/24 [90/2170112] via 209.165.200.226, 00:00:05,
    Serial0/0/0
D   10.1.2.0/24 [90/2170112] via 209.165.200.226, 00:00:05,
    Serial0/0/0
 192.168.10.0/24 is variably subnetted, 2 subnets, 3 masks
C   192.168.10.0/24 is directly connected, GigabitEthernet0/0
L   192.168.10.1/32 is directly connected, GigabitEthernet0/0
 192.168.11.0/24 is variably subnetted, 2 subnets, 3 masks
C   192.168.11.0/24 is directly connected, GigabitEthernet0/1
L   192.168.11.1/32 is directly connected, GigabitEthernet0/1
209.165.200.0/24 is variably subnetted, 2 subnets, 3 masks
C   209.165.200.224/30 is directly connected, Serial0/0/0
L   209.165.200.225/32 is directly connected, Serial0/0/0
R1#
```

- When a packet arrives at a router destined for a remote network, it will send the packet to the next hop address corresponding to the destination network address in its routing table.
- For example, if the R1 router in the figure to the left receives a packet destined for a device on the 10.1.1.0/24 network, it will send it to the next hop address of 209.165.200.226.
- Notice in the routing table, a default gateway address is not set – if the router receives a packet for a network that isn't in the routing table, it will be dropped.

8.3 Routers

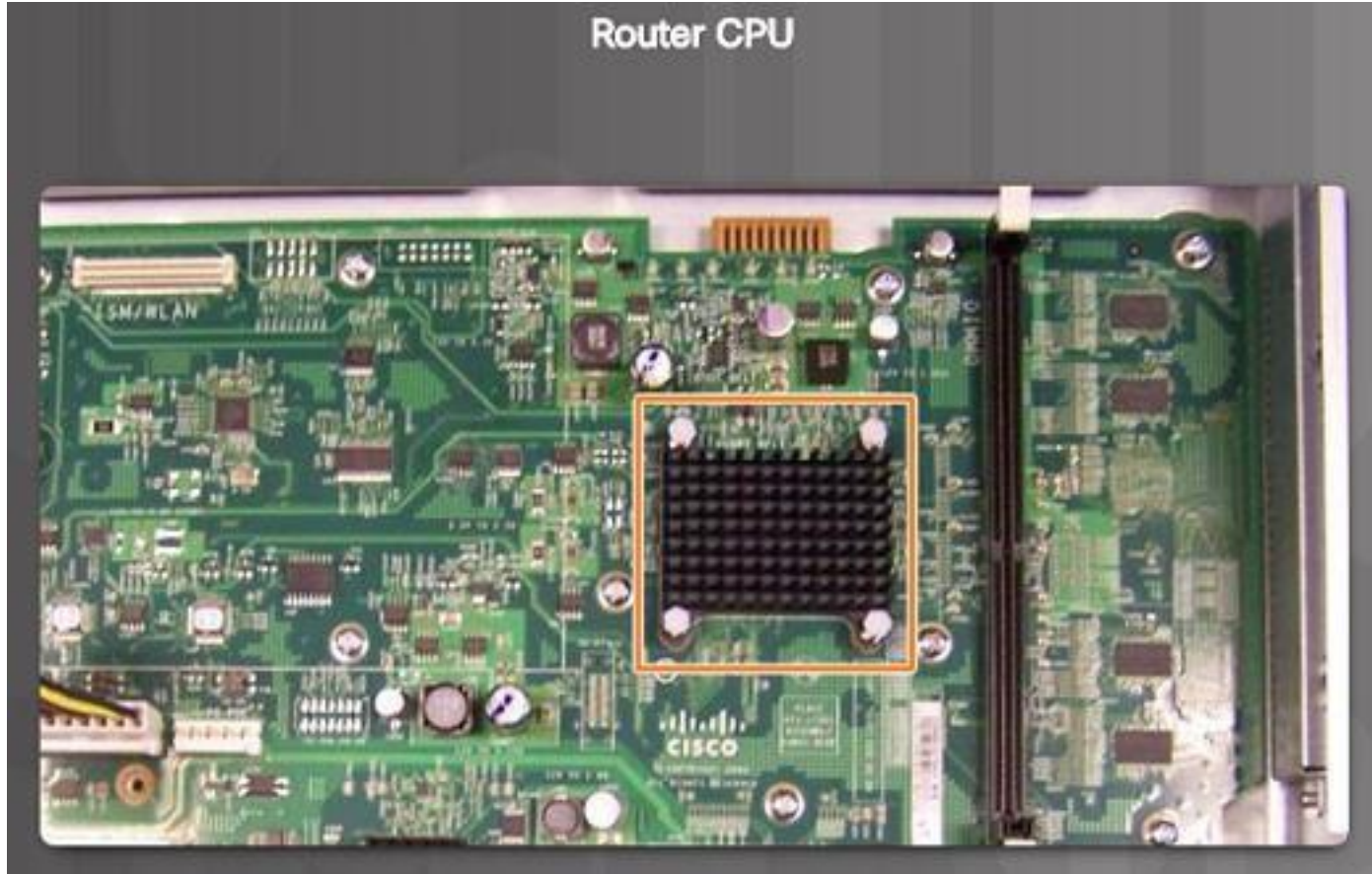
Anatomy of a Router

A Router is a Computer



- **A router is a computer.** Like computers, a **router requires a CPU, an operating system, and memory.**
- Cisco routers are designed to meet the needs of wide variety of businesses and networks:
 - Branch – Teleworkers, small businesses, and medium-size branch sites.
 - WAN – Large businesses, organizations and enterprises.
 - Service Provider – Large service providers.
- The focus of the CCNA certification is on the branch family of routers.

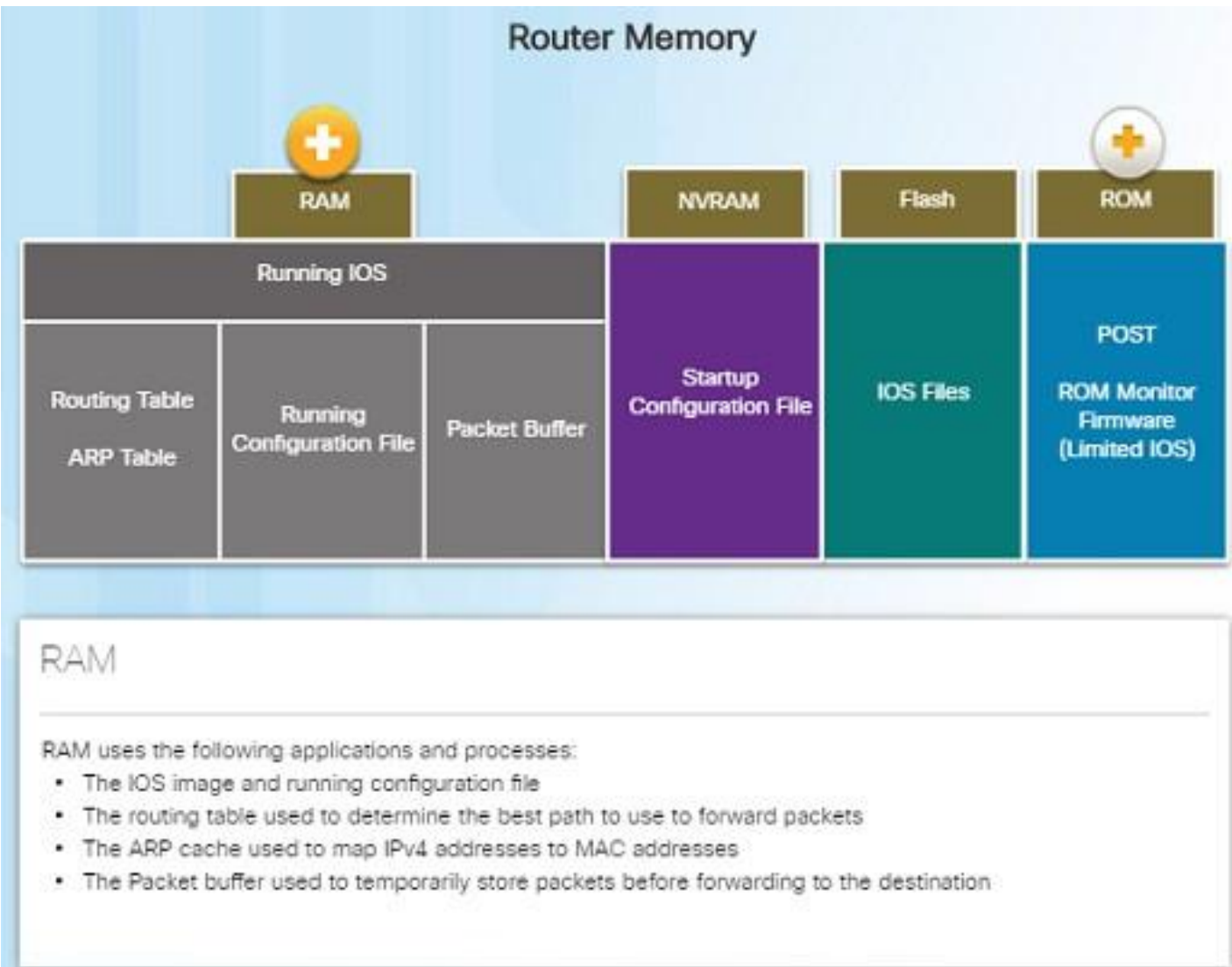
Router CPU and OS



- Like computers, Cisco routers require a CPU to execute OS instructions including system initialization, routing functions and switching functions.
- The component highlighted in the figure to the left is **the CPU of a Cisco 1941** with the heatsink attached. A heatsink is used to dissipate the heat from the CPU for cooling purposes.
- The CPU requires an operating system to provide routing and switching functions. **Most Cisco devices use the Cisco Internetwork Operating System (IOS).**

Anatomy of a Router

Router Memory



- Volatile memory – requires continual power to store information.
- Non-volatile memory – does not require continual power.
- A router uses four types of memory:
 - RAM – Volatile memory used to store applications, processes, and data needed to be executed by the CPU.
 - ROM – Non-volatile memory used to store crucial operational instructions and a limited IOS. ROM is firmware embedded on an integrated circuit inside of the router.
 - NVRAM – Non-volatile memory used as permanent storage for the startup configuration file (startup-config).
 - Flash – Non-volatile memory used as permanent storage for the IOS and other operating system files such as log or backup files.



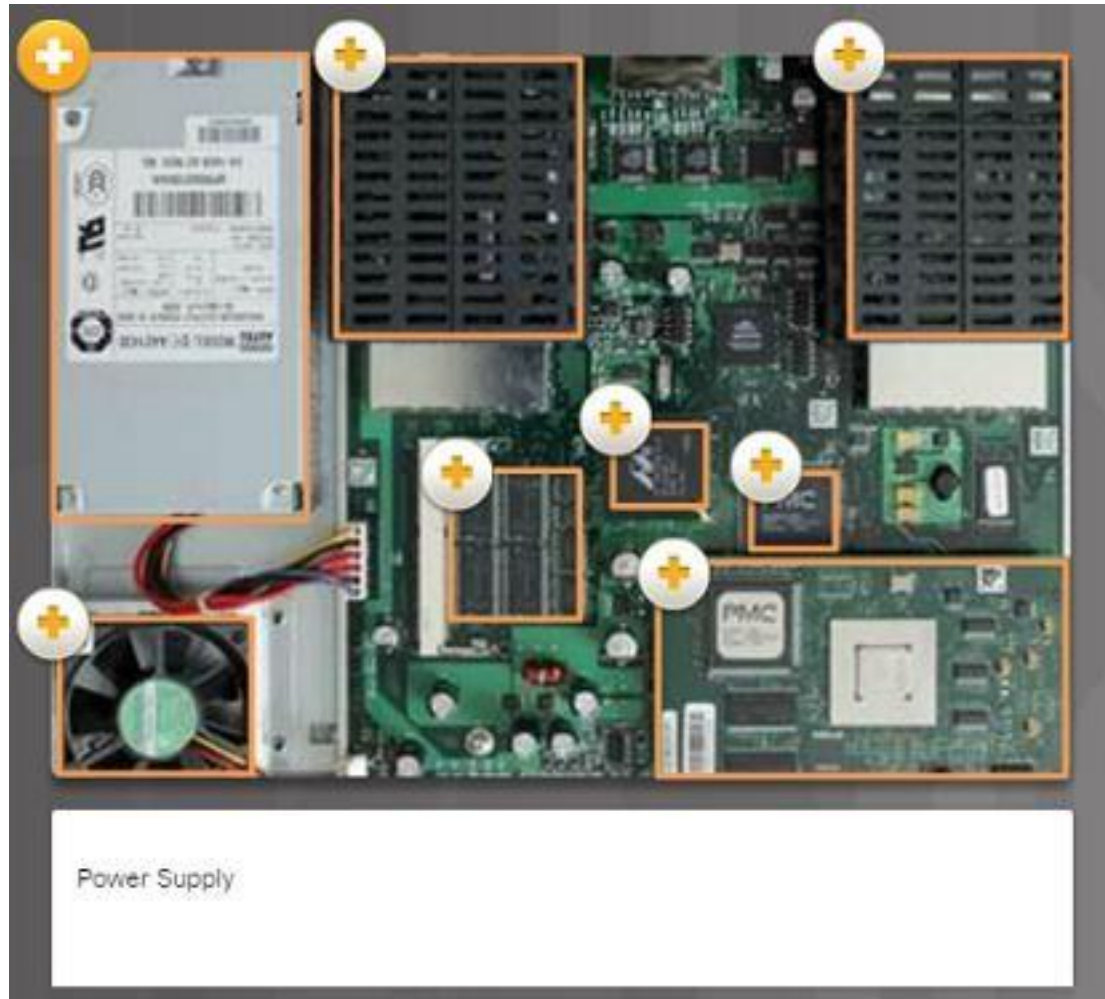
Anatomy of a Router

Router Memory

Memory	Volatile / Non-Volatile	Stores
RAM	Volatile	- Running IOS - Running configuration file - IP routing and ARP tables - Packet buffer
ROM	Non-Volatile	- Bootup instructions - Basic diagnostic software - Limited IOS
NVRAM	Non-Volatile	Startup configuration file
Flash	Non-Volatile	- IOS - Other system files

Anatomy of a Router

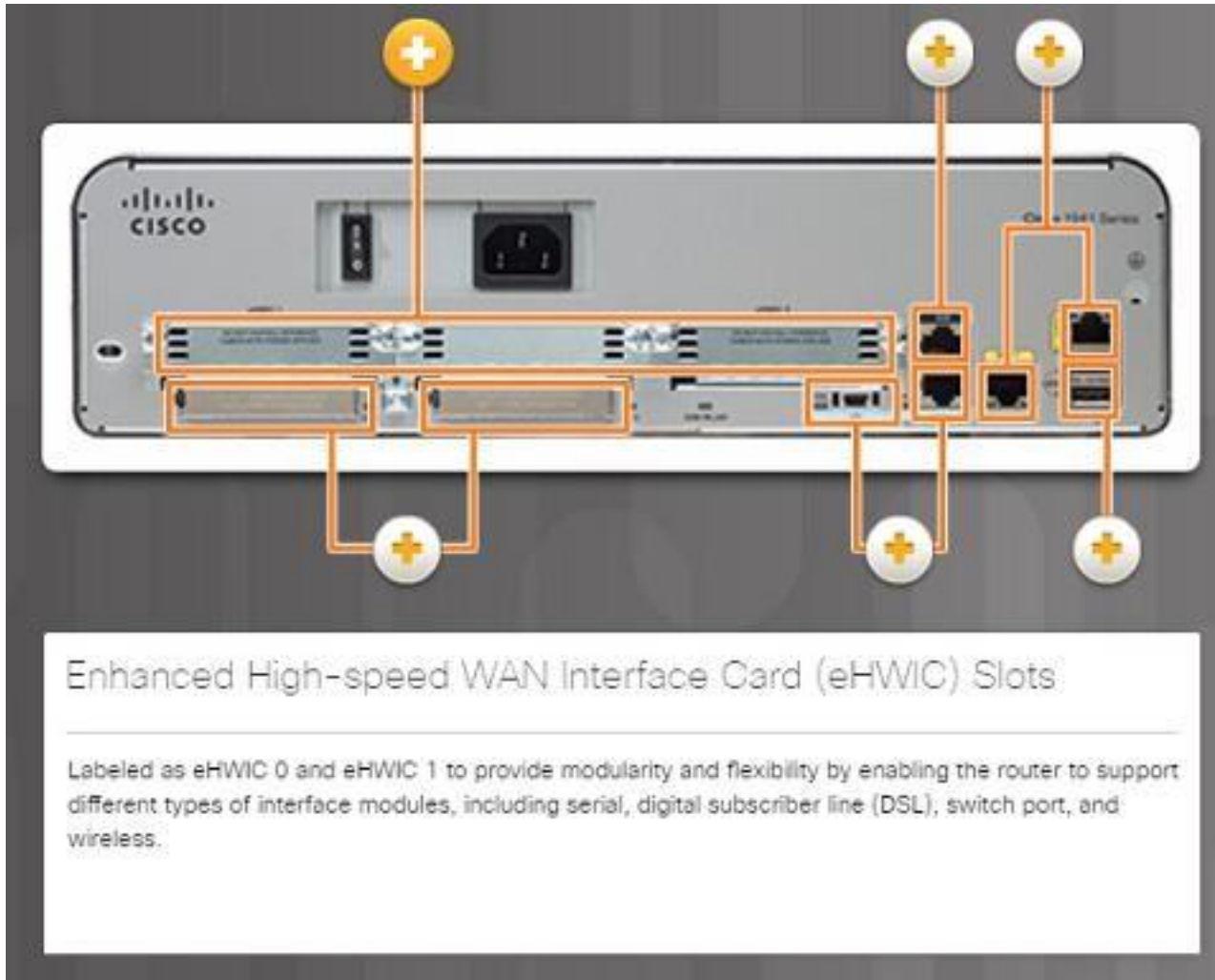
Inside a Router



- There are numerous types and models of routers, however, they all have the same general hardware components:
 - Power supply
 - Cooling fan
 - SDRAM - Synchronous Dynamic RAM
 - Non-volatile RAM (NVRAM)
 - CPU
 - Heat shields
 - Advanced Integration Module (AIM)

Anatomy of a Router

Connect to a Router



- Cisco devices, routers, and switches typically interconnect many devices. The Cisco 1941 router backplane includes the following ports and connections:
- Enhanced High-speed WAN Interface Card (eHWIC) Slots
- Auxiliary (AUX) – RJ-45 port for remote management.
- Console Port – Used for initial configuration and Command Line Interface access – RJ-45 or USB Type-B (mini-B USB)
- Gigabit Ethernet used to provide LAN access by connecting to switches, users, or to other routers.
- Compact Flash Slots – Labeled as CF0 and CF1 and used to provide increased storage flash space upgradable to 4GB.
- USB port – used to provide additional storage space.

LAN and WAN Interfaces

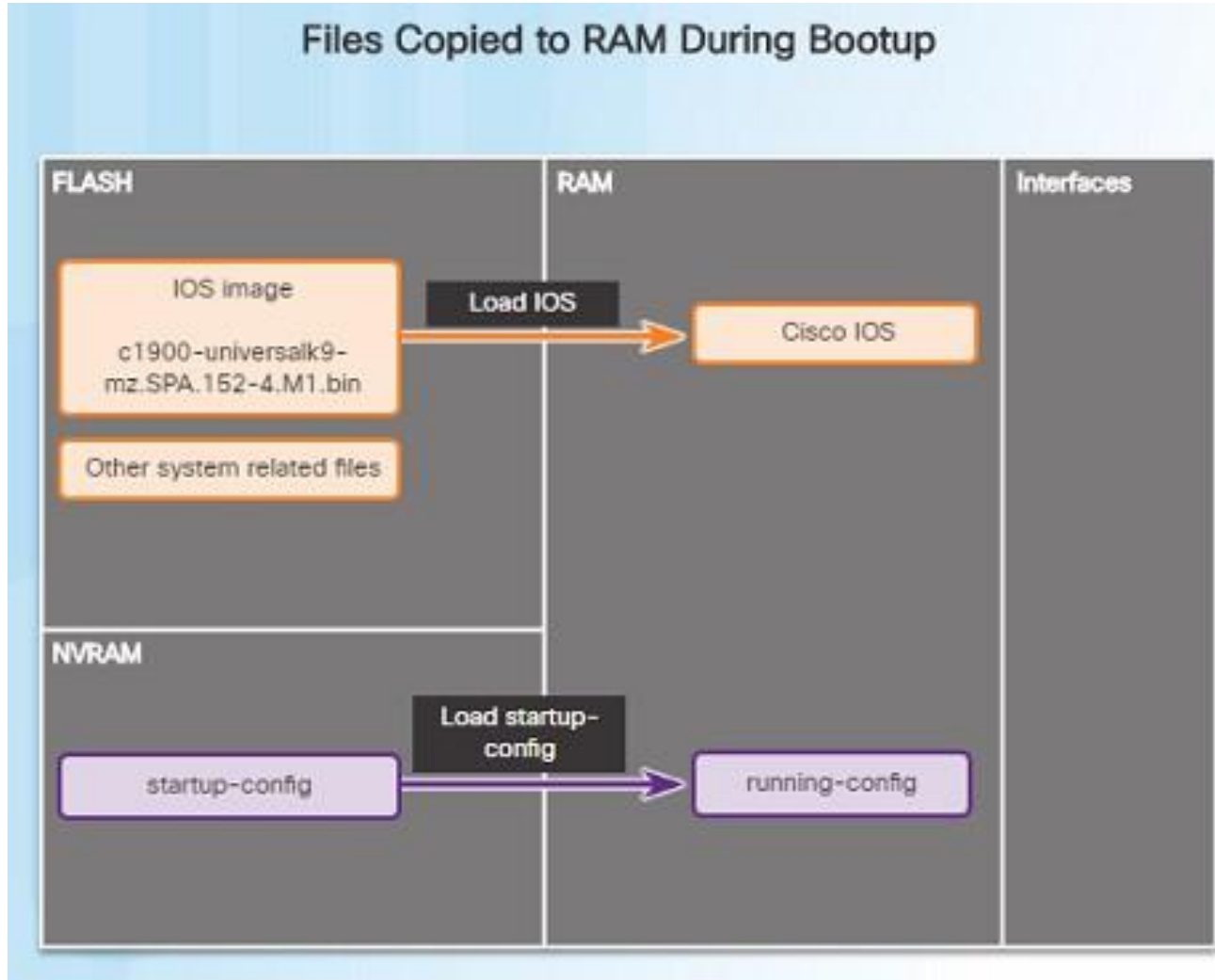
- Cisco router connections can be classified in two categories:
 - In-band router interfaces – LAN and WAN interfaces
 - Management ports – Console and AUX ports



- The most common ways to access user EXEC mode in the CLI environment on a Cisco router:
 - **Console** – This is a physical management port that provides out-of-band access to the Cisco router. Out-of-band means that it is dedicated and does not require network services to be configured on the router.
 - **Secure Shell (SSH)** – This is a secure method of remotely establishing a CLI connection over a network. SSH does require active networking services configured.
 - **Telnet** – Telnet is an insecure method of remotely establishing a CLI session through a virtual interface over a network. The connection is not encrypted.

Router Boot-up

Bootset Files

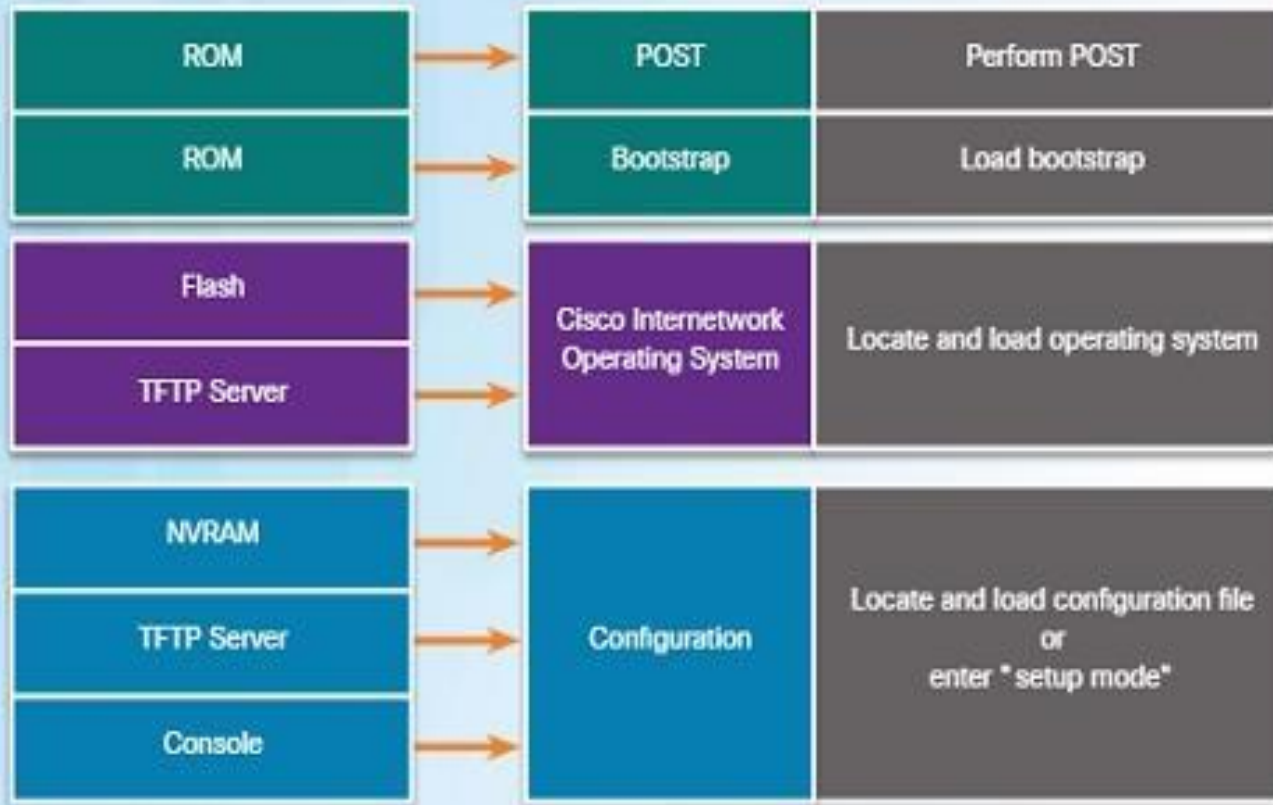


- Cisco routers and switches load the IOS image and startup configuration file into RAM when they are booted.
- The running configuration is modified when the network administrator makes any changes. These changes should be saved to the startup configuration file in NVRAM in order for them to take effect on the next reboot of the router or during in the event of a power loss.

Router Boot-up

Router Bootup Process

How a Router Boots Up

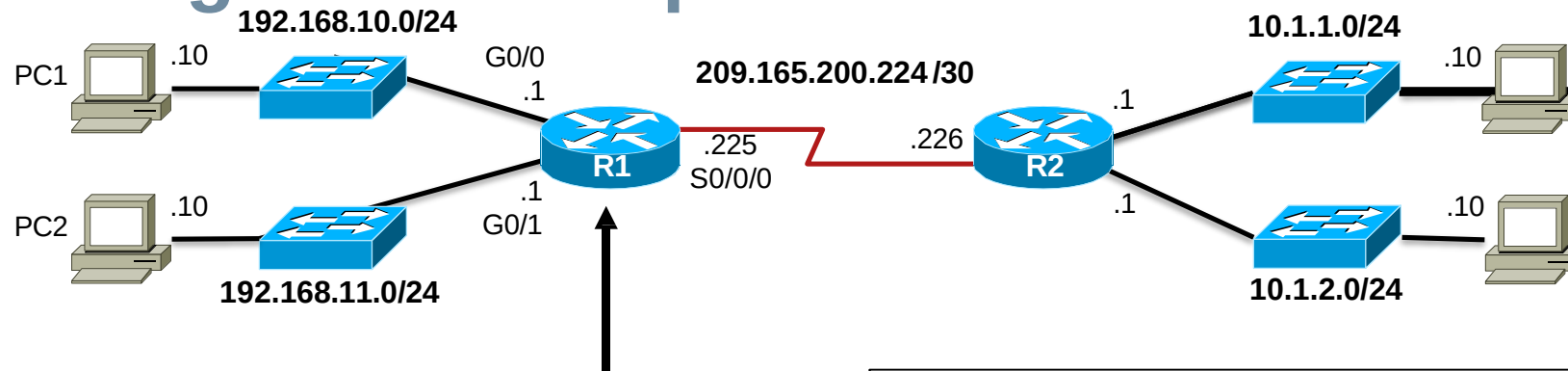


- Three major phases to the bootup process of a router:
- Perform the POST and load the bootstrap program – During the Power-on Self-Test, the router executes diagnostics from ROM on various hardware components. After the POST, the bootstrap program is copied from ROM into RAM and its job is to locate the Cisco IOS and load it into RAM.
- Locate and load the Cisco IOS software – Typically, the IOS is stored in flash memory and is copied into RAM for execution by the CPU.
- Locate and load the startup configuration file or enter setup mode – The bootstrap program then copies the startup config file from NVRAM into RAM and becomes the running configuration.

Configure a Cisco Router

Configure Initial Settings

Router Configuration Steps



```
Router> enable
Router# configure terminal
Enter configuration commands, one per line.
End with CNTL/Z.
Router(config)# hostname R1
R1(config)#
```

OR

```
Router> en
Router# conf t
Enter configuration commands, one per line.
End with CNTL/Z.
Router(config)# ho R1
R2(config)#
```

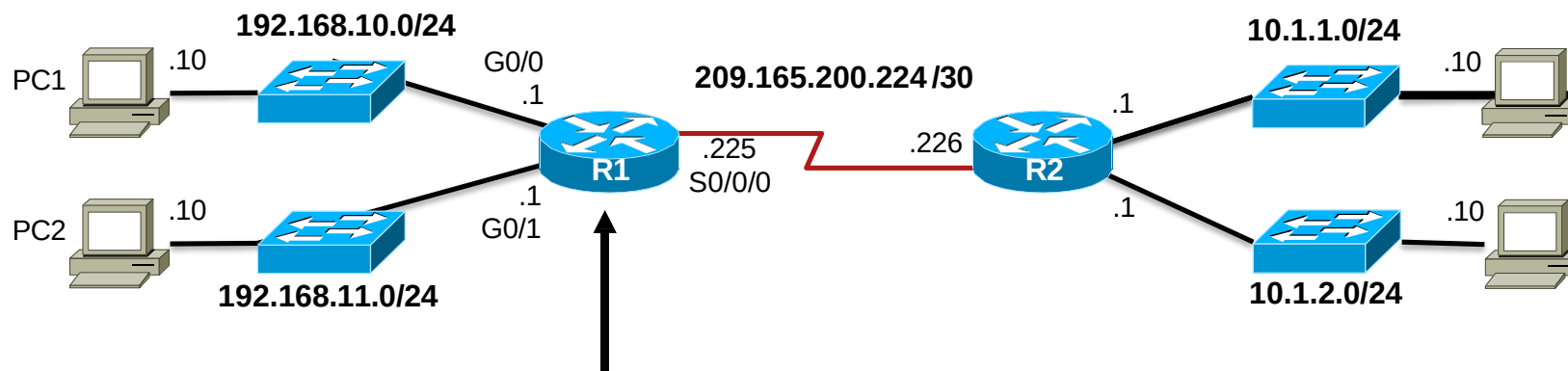
```
R1(config)# enable secret class
R1(config)#
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# exit
R1(config)#
R1(config)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# exit
R1(config)#
R1(config)# service password-encryption
R1(config)#
```

```
R1(config)# banner motd #
Enter TEXT message. End with the character '#'.
*****
WARNING: Unauthorized access is prohibited!
*****
#
R1(config)#
```

```
R1# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

Configure Interfaces

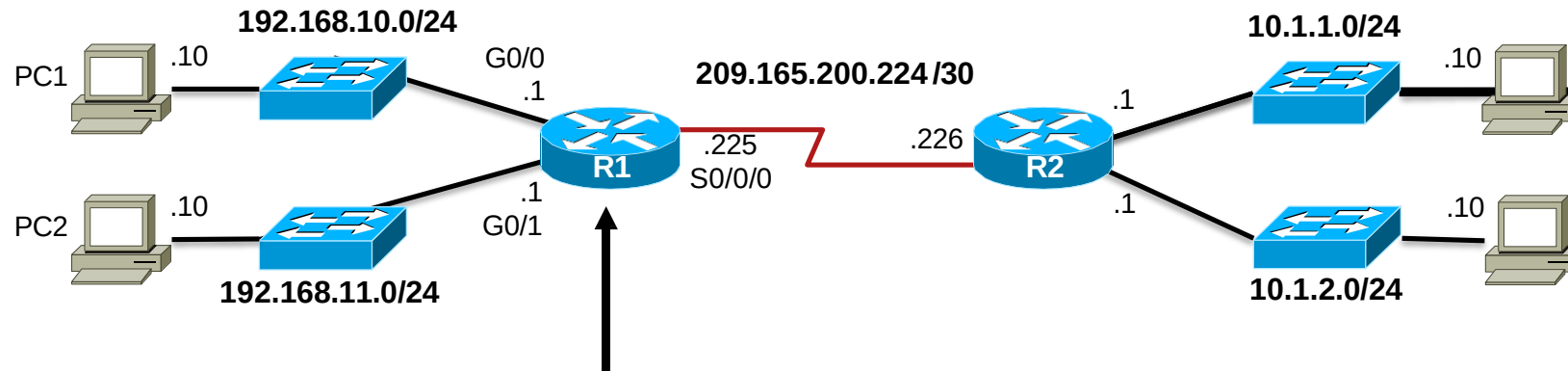
Configure LAN Interfaces



```
R1# conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)# interface gigabitethernet 0/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# description Link to LAN-10
R1(config-if)# no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up
R1(config-if)# exit
R1(config)#
R1(config)# int g0/1
R1(config-if)# ip add 192.168.11.1 255.255.255.0
R1(config-if)# des Link to LAN-11
R1(config-if)# no shut
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up
R1(config-if)# exit
R1(config)#
```

Configure Interfaces

Verify Interface Configuration

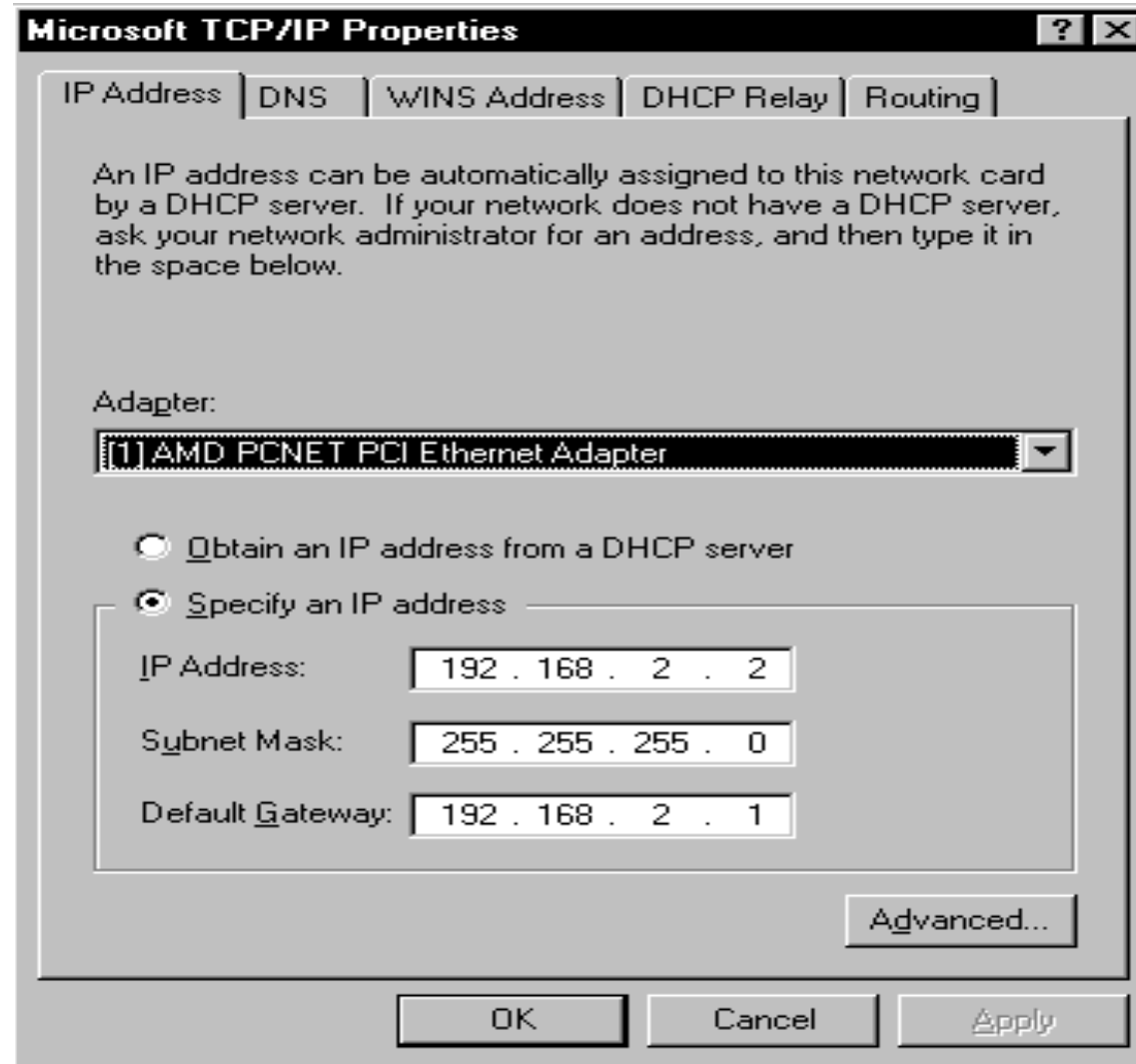


```
R1# show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.10.1    YES manual  up          up
GigabitEthernet0/1       192.168.11.1    YES manual  up          up
Serial0/0/0               209.165.200.225 YES manual  up          up
Serial0/0/1               unassigned      YES NVRAM   administratively down down
Vlan1                     unassigned      YES NVRAM   administratively down down
R1#
R1# ping 209.165.200.226

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.226, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/9 ms

R1#
```

Configuring the Default Gateway



Microsoft TCP/IP Properties

IP Address | DNS | WINS Address | DHCP Relay | Routing

An IP address can be automatically assigned to this network card by a DHCP server. If your network does not have a DHCP server, ask your network administrator for an address, and then type it in the space below.

Adapter:

☐ Obtain an IP address from a DHCP server
☒ Specify an IP address

IP Address:
 Subnet Mask:
 Default Gateway:

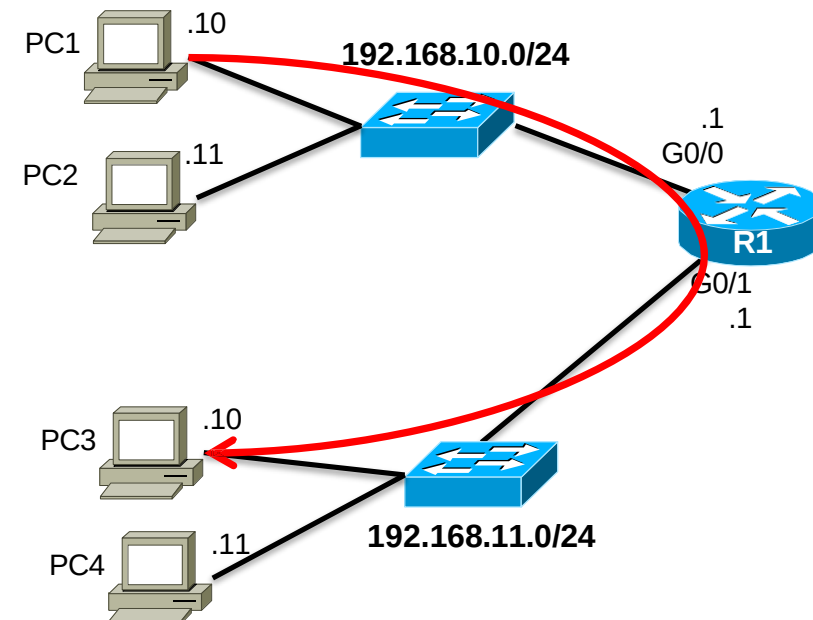
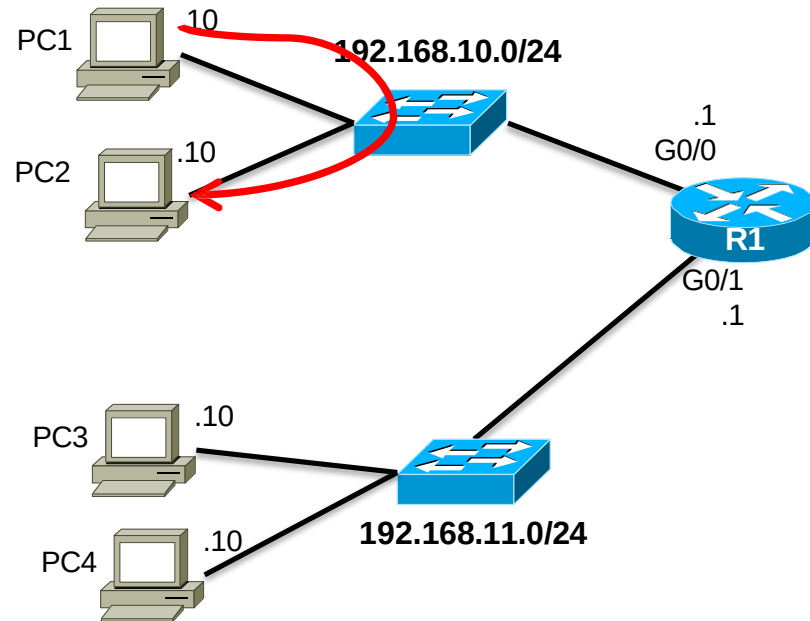
Advanced...

OK Cancel Apply



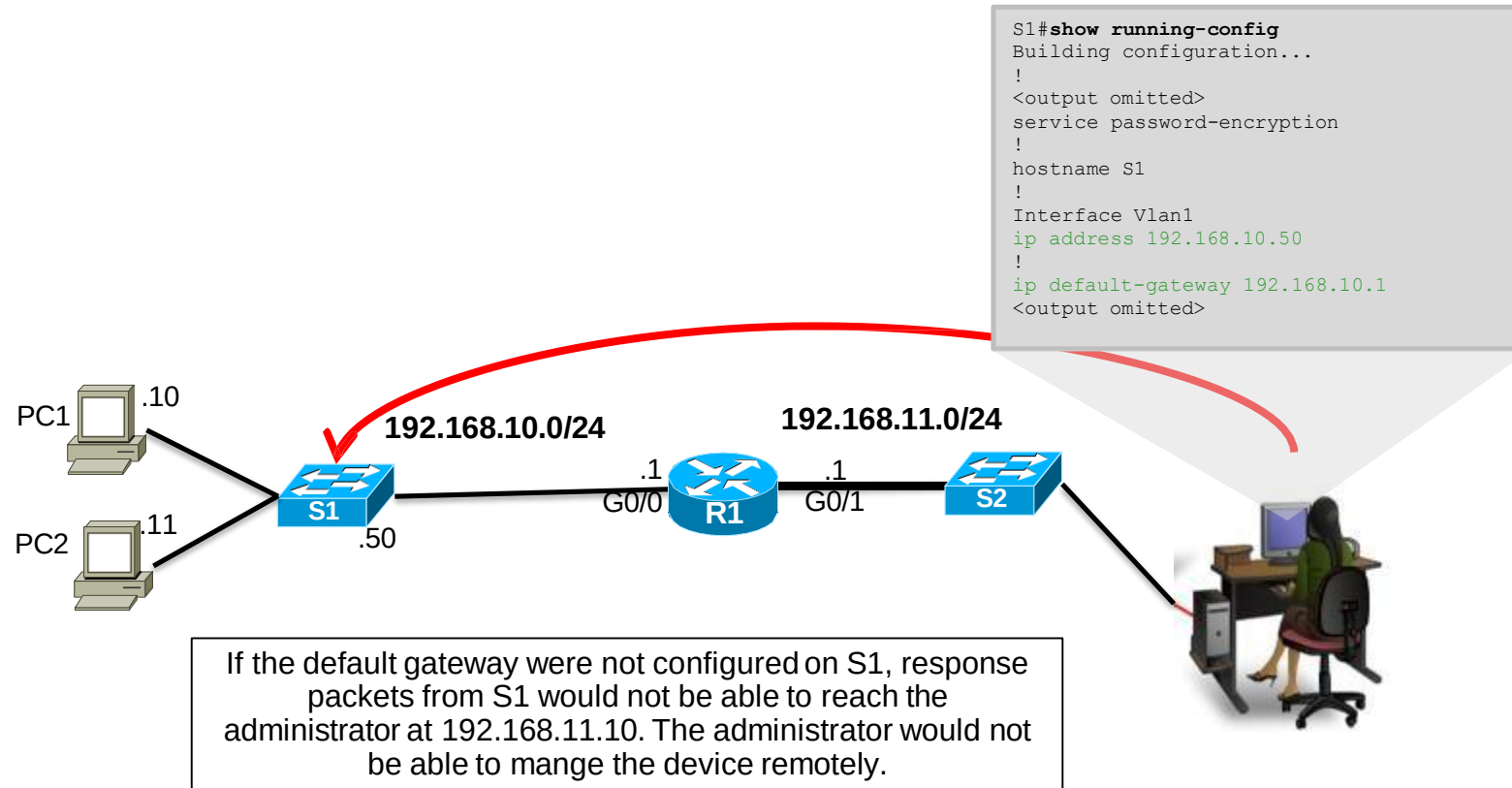
Configuring the Default Gateway

Default Gateway on a Host



Configuring the Default Gateway

Default Gateway on a Switch





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